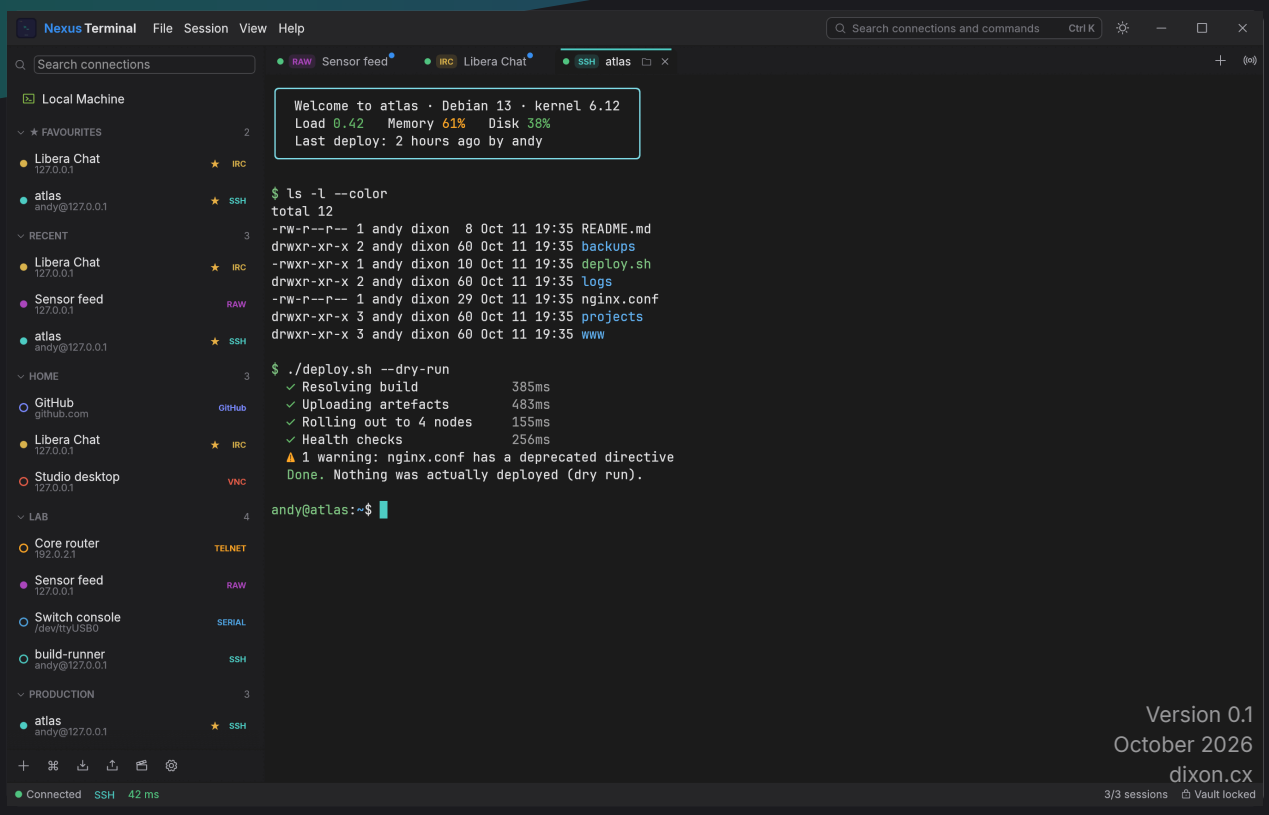
THE USER GUIDE

Nexus Terminal

Every terminal you need. One window.
Zero faff.

SSH · Telnet · RAW TCP · Serial · VNC · IRC · GitHub · Local shells



Version 0.1
October 2026
dixon.cx

Start here.

(Yes, even if you never read manuals. *Especially* then.)

This guide is built for skimming. You do **not** have to read it in order, cover to cover, like it's homework. Find the bit you need, grab it, get back to work. Every chapter opens with the time it takes and what you'll be able to do afterwards, and a **TL;DR** if you only want the gist.

I just want to…

…get connected to something *right now*	Five-minute start, p. 2
…work out what all the buttons do	The grand tour, p. 7
…SSH into a server (keys, jump hosts, 2FA)	SSH, p. 13
…talk to a router or Arduino over a USB cable	Serial, p. 16
…copy files to and from a server	Files, p. 35
…type the same thing into ten servers at once	Never type it twice, p. 32
…keep my passwords safe	Passwords & the vault, p. 40
…move over from Alchementary Link	Moving house, p. 50
…fix something that's broken	When things go wrong, p. 53
…just see the keyboard shortcuts	The cheat sheet, p. 57

The little boxes

TL;DR
The whole chapter in three lines.

Heads up
A thing that might trip you up.

Quick win
Something you can try in under a minute.

Careful
Could lose data or weaken security. Read it.

Pro tip
A trick that saves you time later.

Side quest · optional, skip it if you're in a hurry
Interesting, but not needed to get the job done.

Shortcuts: `Ctrl / ⌘` means `Ctrl` on Windows and Linux, `⌘` (Command) on a Mac. So `Ctrl / ⌘` + `K` is `Ctrl` + `K` or `⌘` + `K` depending on your machine.

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1

PART 1

Get going

From zero to connected before your tea goes cold.

- Five-minute start
- The grand tour
- Profiles: your address book

GET GOING · CHAPTER 1

Five-minute start

about **5 minutes** You'll be connected to something and know the one shortcut that matters

TL;DR

Install it, open it, press `Ctrl / ⌘` + `N`, fill in a host, hit **Create & Connect**. Done. Lost? Press `Ctrl / ⌘` + `K` and type what you want.

1.1 Install it

Grab the zip for your machine from `dixon.cx/downloads/nexus`. They're named `NexusTerminal-<os>-<arch>.zip`, so a modern Mac wants `darwin-arm64` and most PCs want `amd64`.

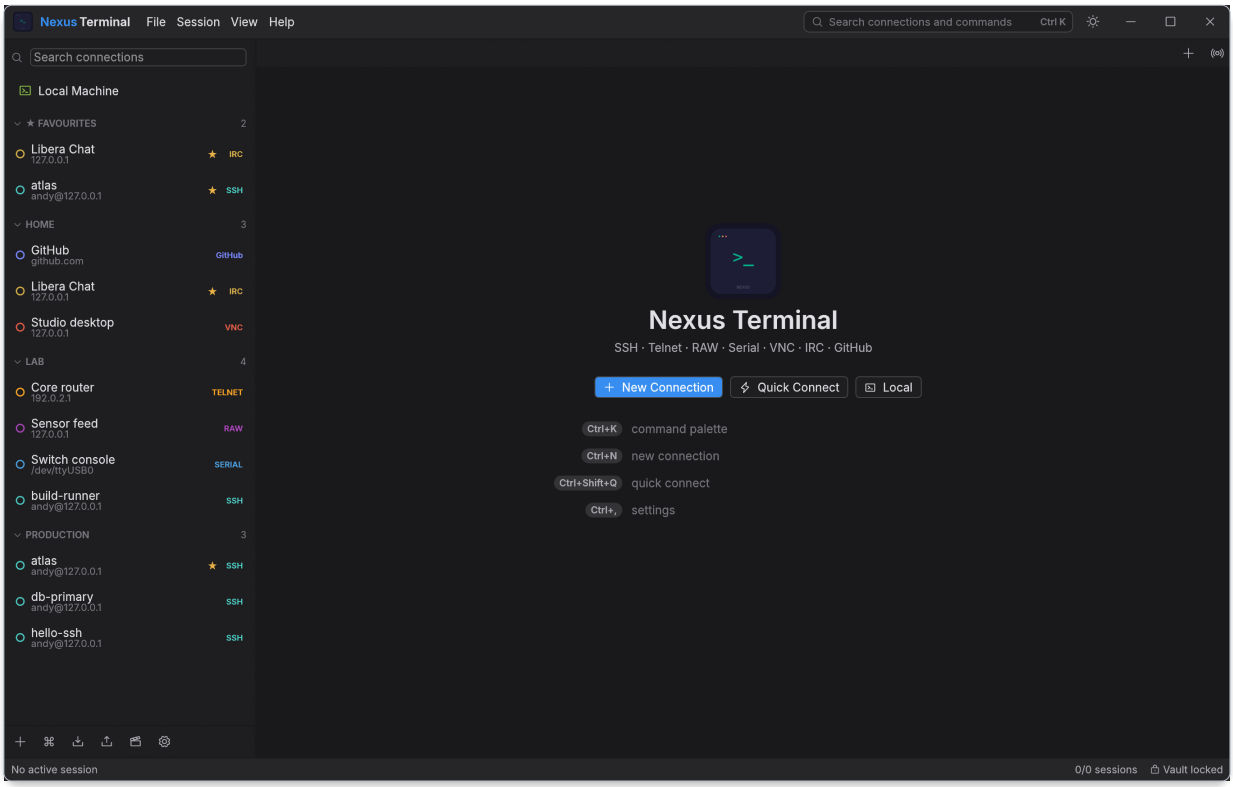
- **Windows** Unzip it and run `NexusTerminal-windows-amd64.exe`. No installer, no admin rights, nothing in the registry. Pop it somewhere sensible like `C:\Tools` and pin it to the taskbar.
- **macOS** Unzip it and drag `NexusTerminal-darwin-arm64.app` into **Applications**. If macOS grumbles that it can't check the developer, right-click the app, choose **Open**, then **Open** again. You only have to do that once.
- **Linux** Unzip it and run `./install.sh`. It copies `nexus-terminal` into `/.local/bin` and adds a launcher entry with the icon, so it turns up in your app menu. (Make sure `/.local/bin` is on your `PATH`.)

Build it yourself · optional, skip it if you're in a hurry

It's a single Rust program. With Rust installed: `cd rust && cargo run --release`. The finished binary lands in `rust/target/release/nexus-terminal`. The build guide in `docs/build-and-release.md` has the full story.

1.2 Open it

The first time you open Nexus Terminal you get a calm, empty welcome screen. Three buttons, four shortcuts. That's it.



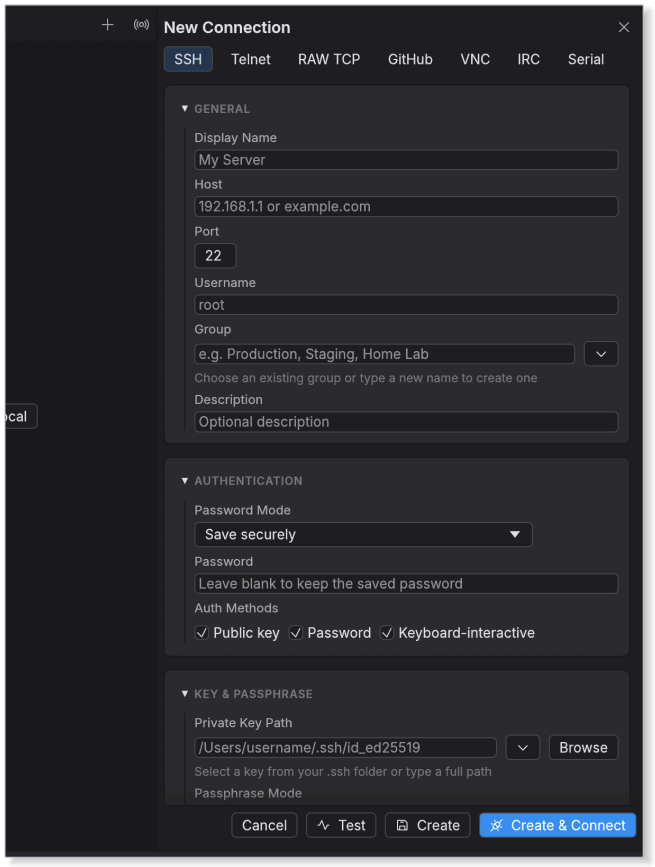
The welcome screen. Your saved connections will live down the left; here there are a few already.

> ⚠ **Coming from Alchementary Link?**
>
> Nexus Terminal spots your old data on first run and copies your profiles, vault and recordings across automatically. You'll see a little toast saying so. Your old data is left exactly where it was. The details are in Moving house on p. 50.

1.3 Make your first connection

1. Click **New Connection**, or press `Ctrl / ⌘` + `N`.
2. Pick the protocol along the top: SSH, Telnet, RAW TCP, GitHub, VNC, IRC or Serial.
3. Fill in the **General** bit: a **Display Name** you'll recognise, the **Host**, the **Port** and your **Username**.
4. Click **Create & Connect**.

That's genuinely it. A tab opens, you log in, and next time it's one click in the sidebar.



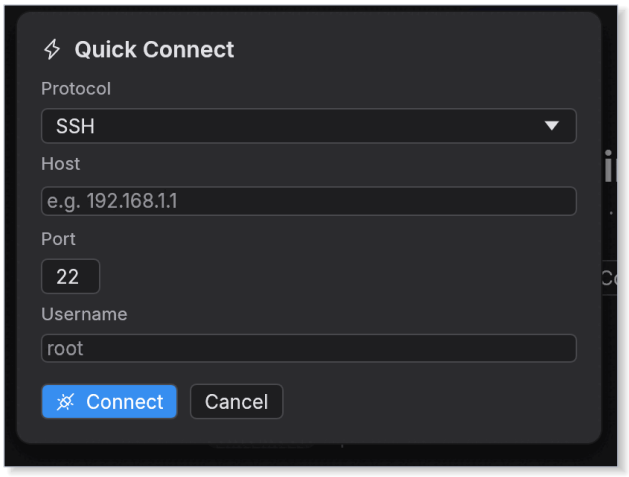
A brand-new SSH profile. Only the top few fields are needed to get going.

> **Quick win**
>
> Click **Test** before you save. Nexus Terminal checks the host and port are reachable and tells you the round-trip time. If that fails, no amount of password-fiddling will help, so it's a great first check.

1.4 Two shortcuts for impatient people

→ Quick Connect

Press `Ctrl / ⌘` + `Shift` + `Q`. For a one-off session you don't want to save. Pick SSH, Telnet, RAW, IRC or Serial, type the host (or serial device), hit `Enter`. Nothing gets saved.

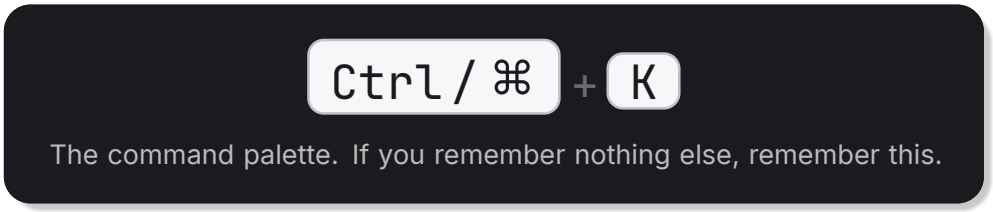


Quick Connect. In, out, nobody gets hurt.

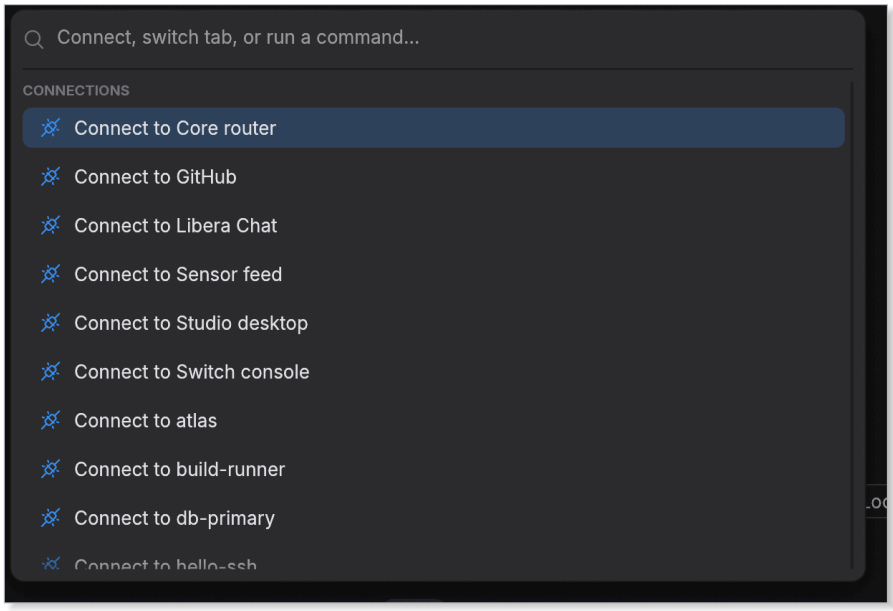
→ Local terminal

Click **Local Machine** at the top of the sidebar (or **Local** on the welcome screen) for a shell on *this* computer. When you type `exit`, the tab says `[Exited - press enter to restart]`. Press `Enter` and you've got a fresh shell in the same tab.

1.5 The one shortcut to rule them all



Type a few letters of *anything*: a connection name, a tab, "theme", "split", "lock", "snippets". Use `↑` `↓` to pick, `Enter` to go, `Esc` to bail. It's fuzzy, so "atl" finds *atlas* and "sft" finds the *SFTP Browser*.



The command palette: every connection, tab and action in one searchable list.

Your five-minute checklist

- [ ] Installed and opened Nexus Terminal
- [ ] Opened a **Local Machine** tab and typed `exit`, then `Enter` to get it back
- [ ] Made a profile with `Ctrl` / `⌘` + `N` and clicked **Test**
- [ ] Connected to it
- [ ] Opened the palette with `Ctrl` / `⌘` + `K` and found something by typing three letters

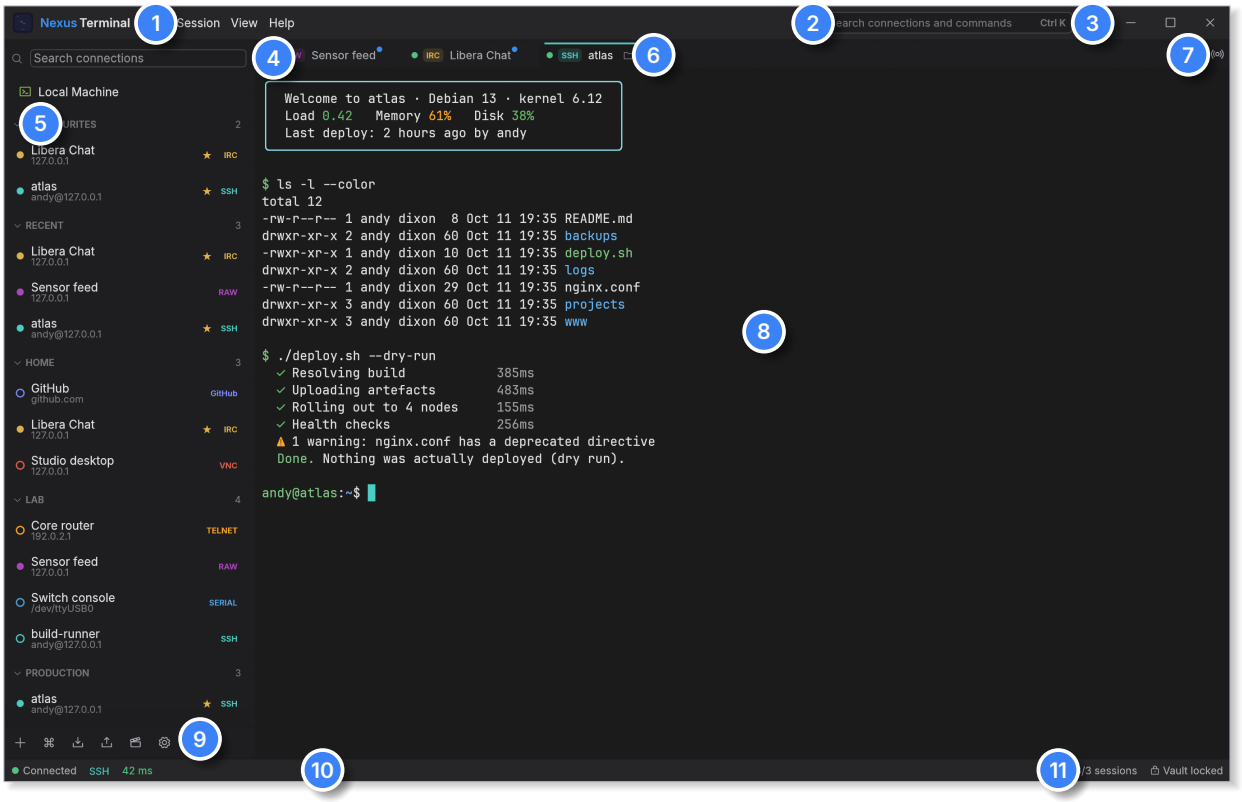
All ticked? You're officially more productive than you were five minutes ago. 🎉

The grand tour

about **4 minutes** | You'll know what every bit of the window is for

TL;DR

Connections on the left. Sessions in tabs across the top. The terminal in the middle. Status along the bottom. Everything else is one `Ctrl / ⌘` + `K` away.



1 **Menus** **File**, **Session**, **View** and **Help**. Drag the title bar to move the window; double-click it to maximise.

2 **Palette** The search box is the command palette. Click it or press `Ctrl / ⌘` + `K`.

3 **Light/dark** Flip between the dark and light themes in one click.

4 **Search** Fuzzy-search your connections by name, host, tag or protocol.

5 **Sections** **Local Machine**, **Favourites** ★, **Recent** (your last five), then one section per group. Click a header to fold it away.

6 **Tabs** One per session, with a coloured state dot, protocol badge and close button. SSH tabs get a little 🗁 for files. A blue dot means "new output while you weren't looking".

7 **New & broadcast** **+** opens a new connection; (◉) toggles broadcast mode.

8 **Workspace** Your terminal (or IRC chat, VNC desktop, GitHub list…). Split it into panes, too.

9 **Footer** New connection, palette, import, export, recordings and settings.

10 **Status** State, protocol and live latency for SSH.

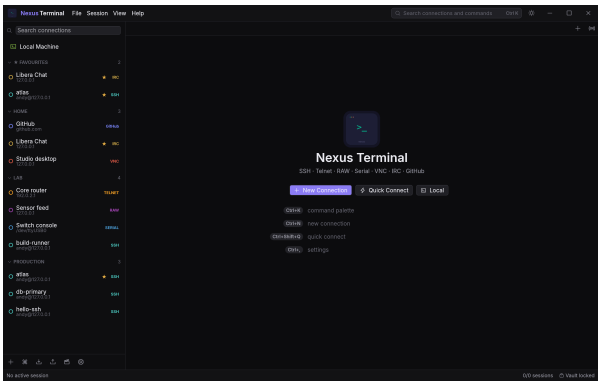
Shows **REC** while recording and **BROADCAST** when broadcasting.

11 **Sessions & vault** How many sessions are connected and whether the vault is locked. Click the vault to open security settings.

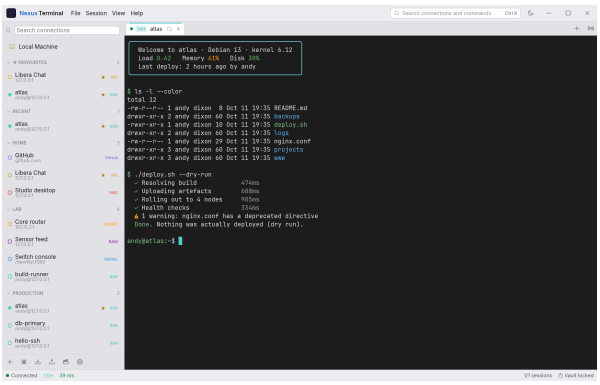
> **Pro tip**
>
> Drag the right-hand edge of the sidebar to make it wider or narrower. Toasts (the little pop-up messages in the top-right) tidy themselves away after a few seconds; click × if you're impatient.

2.1 Three looks

Nexus Terminal comes in **Pro** (the default, a soft charcoal), **Midnight** (darker, bluer, for 2 a.m. deploys) and **Light** (for sunny days and projectors). Switch with the button in the title bar, the **View** menu, or type "theme" into the palette.



Midnight



Light, with a session open

The *terminal* has its own colour schemes too: eleven of them, including **Dracula**, **Nord**, **Tokyo Night**, **Catppuccin Mocha** and **Rosé Pine**. Those live in **Settings** → **Terminal** and the **View** menu.

Profiles: your address book

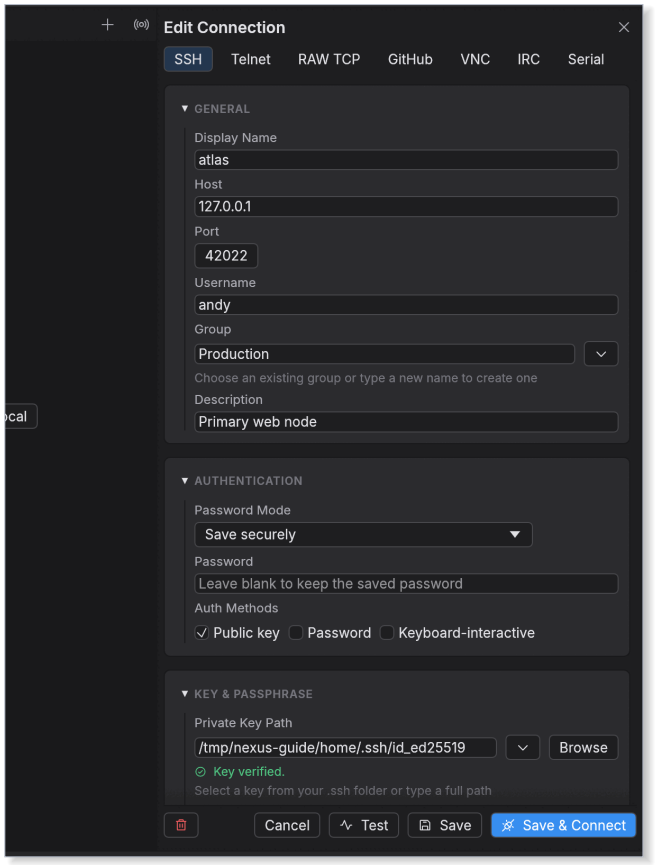
about **6 minutes** | You'll keep dozens of connections tidy without thinking about it

TL;DR

A *profile* is a saved connection. Give it a name and a group, star the ones you use daily, and right-click for everything else. Ctrl-click to select lots at once.

3.1 Create, edit, copy

- **New:** **+** in the sidebar footer or tab strip, `Ctrl` / `⌘` + `N`, or right-click a group header and choose **New connection in …** (it lands in that group automatically).
- **Edit:** double-click a profile, or right-click → **Edit**. The editor slides in on the right.
- **Copy:** right-click → **Duplicate**. Great for "same as web-01 but web-02".
- **Delete:** right-click → **Delete**. This also deletes the saved passwords for that profile.



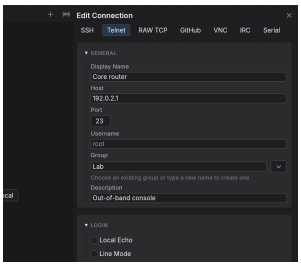
Editing an existing SSH profile. The editor scrolls; sections are grouped so you can ignore most of them.

→ What goes in a profile

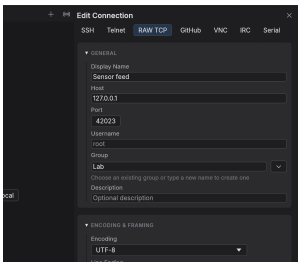
Every profile has the same **General** section, then sections specific to its protocol (Part 2 covers each one). The common bits are:

Display Name	What you'll see in the sidebar. Make it obvious: "db-primary", not "server 3".
Host / Port	Where to connect. Serial profiles have a **Device** instead.
Group	Pick an existing group or type a new name to create one.
Description	One line shown on hover.
Terminal	Font, size (8–36), scrollback (0–100 000 lines) and cursor style, just for this profile.
Tags & Notes	Comma-separated tags (searchable!) and free-form notes for future-you.

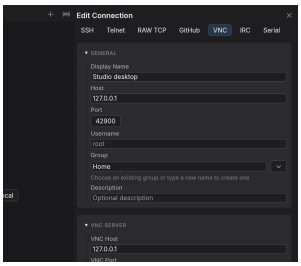
Each protocol adds its own sections underneath. Here are four of them side by side; Part 2 goes through each one.



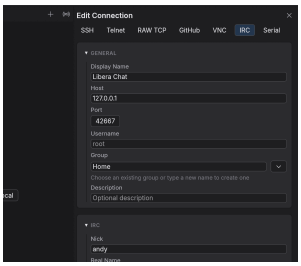
Telnet



RAW TCP



VNC



IRC

> **Tags are secretly brilliant**
>
> Tag things by where they live and what they do (`eu-west`, `db`, `customer-x`). The sidebar search matches tags, so typing "db" instantly narrows hundreds of profiles to the database servers.

3.2 Favourites, groups and Recent

- Right-click → **Add to Favourites** pins a profile into the ★ **Favourites** section at the top.
- **Recent** always shows the last five things you connected to. You don't have to do anything.
- Groups are just the **Group** field. No group? It goes in **Ungrouped**.

3.3 Do lots at once

Hold `Ctrl` (`⌘` on a Mac) and click several profiles. A bar appears at the bottom of the sidebar:

Connect all	opens them all in tabs	**Move group**	moves them (blank = Ungrouped)
Disconnect	closes their sessions	**Delete**	deletes them (asks first)
Tags	add or replace tags	**Clear**	forget the selection

> **Quick win**
>
> Select every web server, choose **Connect all**, then turn on broadcast (`Ctrl` / `⌘` + `Shift` + `B`). You're now typing into all of them at once. Chapter 13 has the details, and the safety advice.

3.4 "It's already open..."

Click a profile that already has a live session and Nexus Terminal asks whether to **Use Existing** (jump to that tab), open a **New Connection** alongside it, or **Cancel**. No more accidental duplicate sessions.

2

PART 2

Connect to all the things

Eight ways in. Pick the one you need; skip the rest.

- SSH: the big one
- Serial: cables and consoles
- Telnet and RAW TCP
- VNC remote desktops
- IRC
- GitHub

SSH: the big one

about **10 minutes (or 2 for the basics)** You'll log in with keys, hop through bastions and never type a 2FA code again

TL;DR

Host, username, key, done. Everything else (jump hosts, proxies, forwards, 2FA, login scripts) is optional and lives in its own fold-away section of the editor. Open only the ones you need.

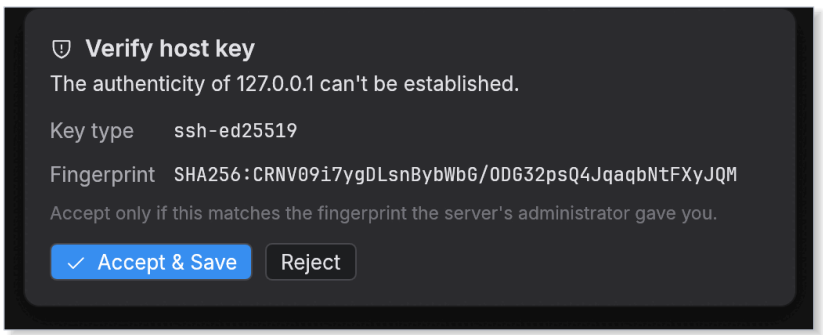
AT A GLANCE

SSH

Default port	22
Logs in with	Password, public key, keyboard-interactive, SSH agent
Extras	SFTP file browser, port forwards, live latency, agent forwarding, TOTP 2FA
Good for	Servers, VMs, Raspberry Pis, anything with `sshd`

4.1 The two-minute version

1. `Ctrl` / `⌘` + `N`, leave the protocol on SSH.
2. **Host**, **Username**. Leave the port on 22 unless you know otherwise.
3. Under **Key & Passphrase**, pick your key from the suggestions (Nexus Terminal lists the keys in your `~/.ssh` folder) and check for a little green **Key verified**.
4. **Create & Connect**. The first time, you'll be asked to trust the server's host key. Say yes if you were expecting it.



First time connecting? You'll be shown the server's fingerprint. **Accept & Save** if it's the server you meant.

> ⚠ **Heads up**
>
> Type the full path to a key if you're not using the suggestions. Nexus Terminal doesn't expand ~ in that box, so write `/home/you/.ssh/id_ed25519` (or `C:\Users\you\.ssh\id_ed25519`).

4.2 How it logs in

Under **Authentication**, tick which **Auth Methods** to try: **Public key**, **Password** and **Keyboard-interactive**. Under **SSH Agent**, tick **Use SSH Agent for Authentication** to use keys already loaded into `ssh-agent` (or Pageant, or the Windows OpenSSH agent).

Every password and passphrase has a *mode*: save it securely, remember it until you quit, or ask every time. That's all explained in Passwords & the vault (p. 40).

> 💡 **Agent forwarding**
>
> Tick **Forward Agent to Remote** (it's `ssh -A`) and you can `git pull` or hop onwards from the server using the keys on *your* machine. Only do it on servers you trust: anyone with root there can use your agent while you're connected.

4.3 Jump hosts: getting through the bastion

Lots of servers hide behind a bastion. Add one or more hops under **Jump Hosts**, each with its own user, host, port and optional key. Nexus Terminal tunnels through them in order, so the chain *you* → *bastion* → *internal-box* → *db-primary* is just three rows.

4.4 Proxies

Stuck behind a corporate proxy? Under **Proxy**, choose **SOCKS5** or **HTTP CONNECT** and fill in the host, port and (optional) username and password. The proxy is used to reach the SSH server, or the first jump host if you have some.

4.5 2FA without the phone faff

If your server asks for a one-time code, paste the Base32 TOTP secret (the text version of the QR code) into **TOTP Secret** and hit **Save**. From then on, Nexus Terminal answers the "Verification code:" prompt itself. **Test** shows today's code so you can check it matches your phone.

> ⚠ **Heads up**
>
> Save the profile first, then save the TOTP secret. And make sure your computer's clock is right: TOTP codes are time-based, so a clock that's a minute out will produce wrong codes.

4.6 Do things automatically on login

Startup A **Startup Command** to run once you're in (say, `tmux attach`), plus **Environment Variables** as `KEY=VALUE`, one per line.

Login Script For old-school boxes that ask questions. Add steps: **Wait for** some text, then **Send** a reply. Repeat as needed.

4.7 Keeping it alive

Under **Advanced**: the **Terminal Type**, the **Keepalive Interval** and **Max Missed Keepalives**, compression, and **Auto-Reconnect** with a maximum number of attempts (1–20). Reconnection backs off politely: 1 s, 2 s, 4 s… up to 30 s.

> **What's that number in the status bar?** · optional, skip it if you're in a hurry
>
> For SSH sessions the status bar shows live latency, the round trip to the server. Green is snappy. If it's in the hundreds and your typing feels sticky, it's the network, not you.

CONNECT · CHAPTER 5

Serial: cables and consoles

about **4 minutes** — You'll be talking to a switch, router or microcontroller over USB

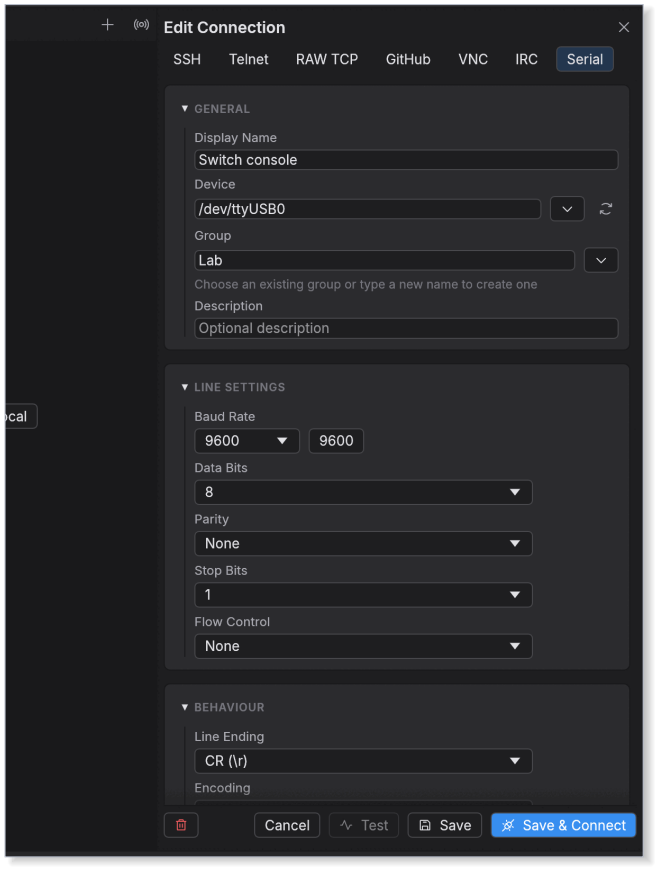
TL;DR

Plug in the adapter, pick the device, match the baud rate. 115200 8N1 is the default and usually right. Right-click the tab to send a break or wiggle DTR/RTS.

Serial

AT A GLANCE

Device	`/dev/ttyUSB0`, `/dev/cu.usbserial-…` or `COM3`
Defaults	115200 baud, 8 data bits, no parity, 1 stop bit, no flow control, CR line endings, DTR and RTS on
Extras	Send Break, toggle DTR/RTS, auto-reconnect when the cable comes back
Good for	Network gear consoles, Arduinos, ESP32s, Raspberry Pi UARTs, 3D printers



A serial profile for a switch console at 9600 baud. Click refresh after plugging in an adapter.

5.1 Setting it up

Device Pick it from the list (hit refresh after plugging in) or type the path.

Line Settings **Baud Rate** (a preset from 300 to 1 000 000, or **Custom**), **Data Bits** (5–8), **Parity** (none, odd, even), **Stop Bits** (1 or 2) and **Flow Control** (none, XON/XOFF, RTS/CTS).

Behaviour **Line Ending**, **Encoding**, **Local Echo** (for devices that don't echo what you type), **Assert DTR on connect**, **Assert RTS on connect** and **Auto-reconnect**.

5.2 While you're connected

Right-click the tab for **Send Break** (handy for getting into a router's ROMMON) and to toggle **DTR** and **RTS** (handy for resetting boards that wire those to reset). Pull the cable out and the session disconnects; with auto-reconnect on, it picks up again when you plug it back in.

Serial tabs are proper terminals, so recording, triggers, broadcast, snippets, find and split panes all work.

> ⚠ **Gibberish on screen?**
>
> Garbled characters nearly always mean the baud rate, parity or stop bits don't match the device. Check the device's manual. 9600 and 115200 are the usual suspects.

> **Linux and macOS quirks**
>
> On Linux, add yourself to the `dialout` group (`uucp` on some distributions) and log out and in again. On a Mac, use the `/dev/cu.*` device, not `/dev/tty.*`.

CONNECT · CHAPTER 6

Telnet and RAW TCP

about **4 minutes** | You'll poke at old kit and raw sockets without reaching for netcat

TL;DR

Telnet is a terminal for old devices. RAW is a packet-level conversation with any TCP port, with a hex view.

6.1 Telnet

AT A GLANCE

Telnet

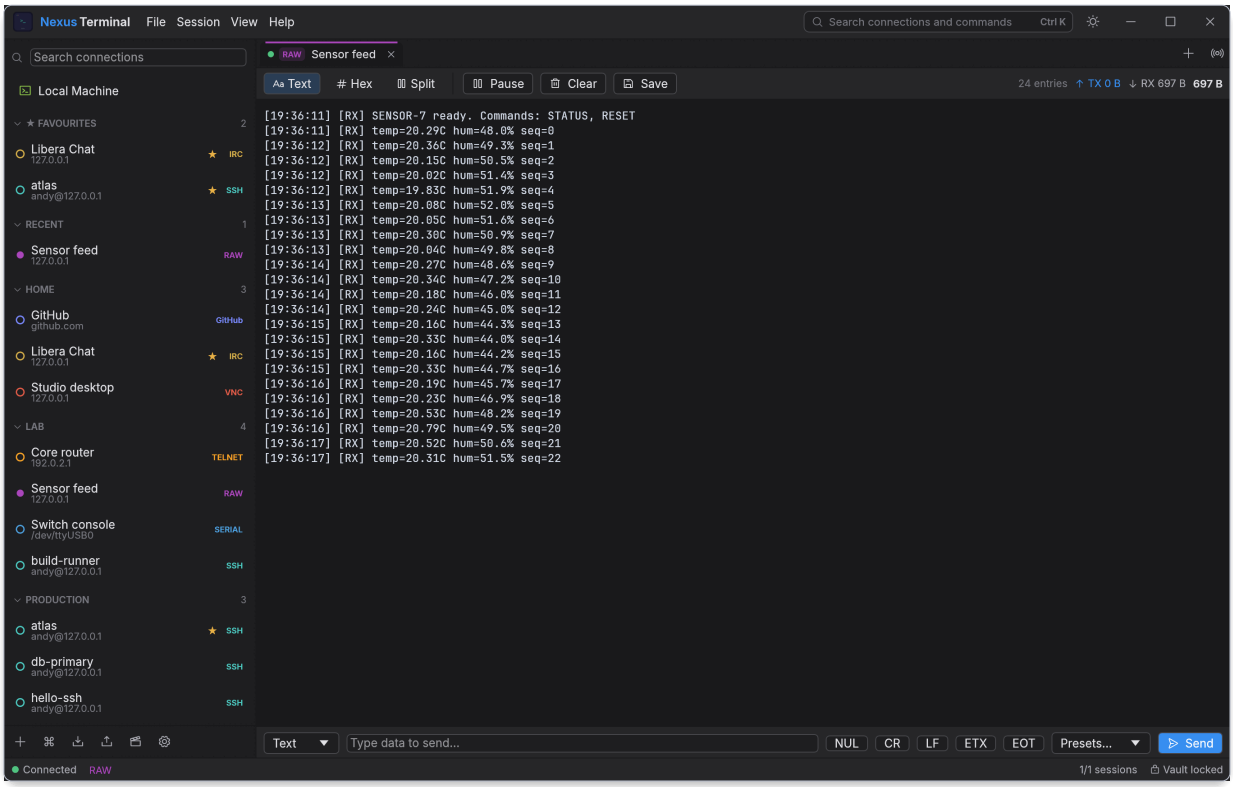
Default port	23
Options	Local echo, line mode, terminal type, encoding (UTF-8, ASCII, ISO 8859-1), line endings, login script
Extras	Send Break from the tab menu, **Debug Negotiation** for when options go wrong

Telnet is plain text

Everything, including your password, crosses the network unencrypted. Fine on a lab bench, not fine over the internet or café Wi-Fi.

6.2 RAW TCP

RAW connects to any TCP port and shows you exactly what goes back and forth. Instead of a terminal you get a traffic log with **Text**, **Hex** and **Split** views.



A RAW session to a chatty sensor, with timestamps and direction markers turned on.

- **Send bar:** pick **Text**, **Hex** or **Escaped** (`\r\n` and friends), type, press `Enter`. One-click buttons send **NUL**, **CR**, **LF**, **ETX** and **EOT**.
- **Presets:** save payloads you send often in the profile under **Payload Presets**, then pick them from **Presets...**.
- **Capture:** **Pause**, **Clear**, and **Save** (writes the whole conversation as JSON).
- **Profile options:** encoding (UTF-8, ASCII, Hex), line ending, timestamps, direction, timeouts and auto-reconnect.

Quick win

Point a RAW profile at a web server on port 80 and send `HEAD / HTTP/1.0\r\n\r\n` in **Escaped** mode. Instant HTTP headers, no browser required.

CONNECT · CHAPTER 7

VNC remote desktops

about **4 minutes** | You'll see and drive a remote desktop, safely

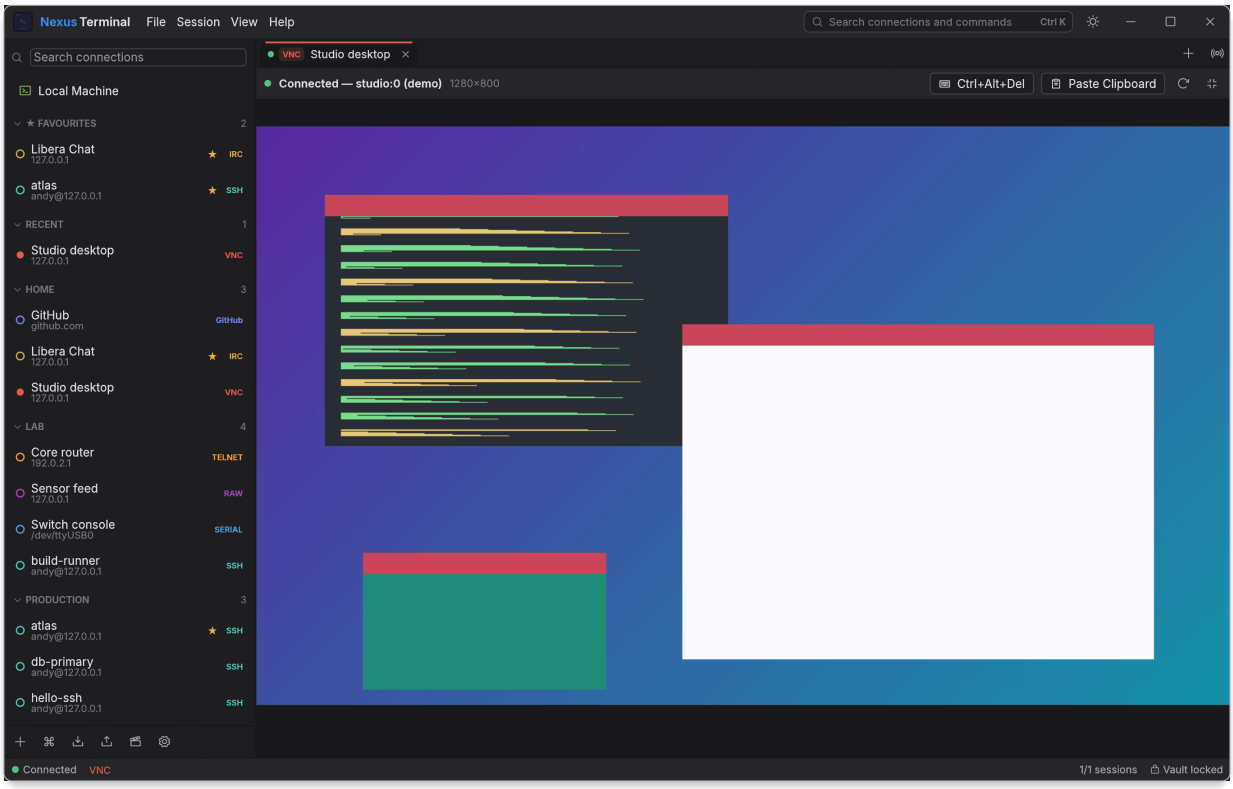
TL;DR

VNC shows you another computer's screen. Tunnel it through SSH (tick a box), because plain VNC isn't encrypted.

AT A GLANCE

VNC

Default port	5900 (plus the display number: display `:1` is 5901)
Display	**View Only**, **Scale to Fit**, **Shared Session**, **Quality (0-9)**
Security	Optional SSH tunnel with password, key or agent login



A remote desktop, scaled to fit the tab.

7.1 The toolbar

Connection state and desktop size on the left. On the right: switch between scaled and 1:1, refresh, **Paste Clipboard** (sends your clipboard to the remote machine) and **Ctrl+Alt+Del**. Click the desktop to give it your keyboard. Anything the remote machine copies lands on your clipboard automatically.

7.2 Tunnel it

Tick **Enable SSH Tunnel** and fill in the SSH host, port, username and login method. The VNC traffic then travels inside an encrypted SSH connection. If you connect to a VNC profile *without* a tunnel, Nexus Terminal warns you and offers **Connect Anyway**, **Edit Profile** or **Cancel**.

> ⚠ **Connecting to a Mac?**
>
> Turn on **VNC viewers may control screen with password** in macOS Screen Sharing settings. Nexus Terminal uses standard VNC password authentication, not Apple's own login method.

CONNECT · CHAPTER 8

IRC

about **3 minutes** You'll be chatting in your project's channels from the same window as your servers

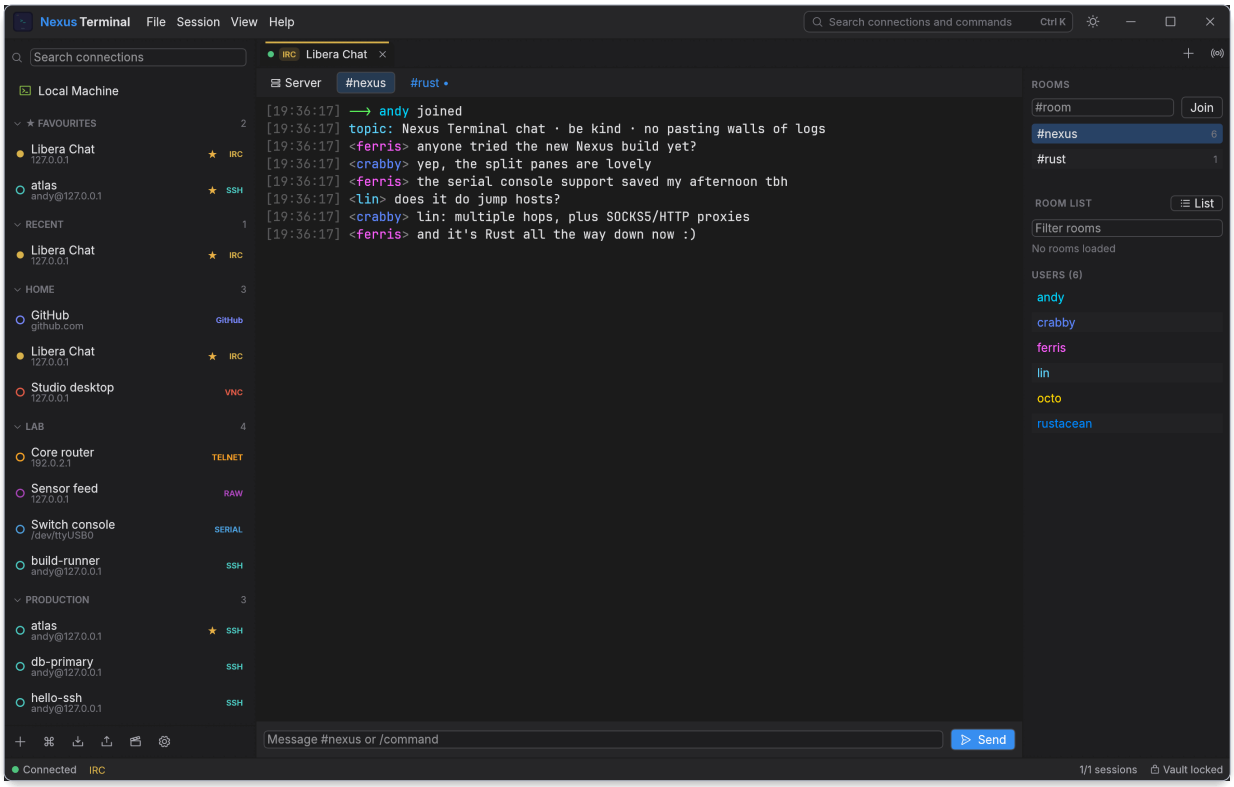
TL;DR

Nick, server, rooms to join. Tabs for each room, `/commands` as you'd expect, `Tab` completes nicks.

AT A GLANCE

IRC

Default port	6667, or 6697 with **Use TLS** (use TLS!)
Extras	NickServ identify on connect, auto-join rooms, room browser, SSH tunnel



Chatting in #nexus. Rooms down the right, users in the room below them.

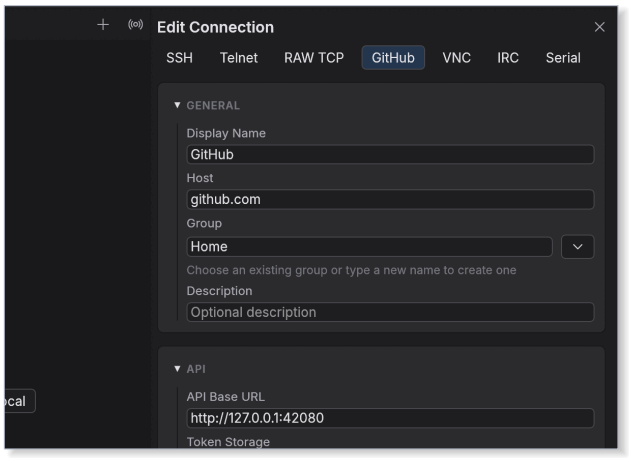
- A **Server** tab plus one per room or private conversation. A dot means unread messages.
- Commands: `/join`, `/part`, `/msg`, `/me`, `/notice`, `/nick`, `/topic`, `/list`, `/names`, `/whois`, `/who`, `/raw`, `/quit` and `/help`.

- `Tab` completes a nick; click a nick to mention them.
- The side panel lists your rooms, a room browser with a filter (click to join) and who’s in the current room.

CONNECT · CHAPTER 9

GitHub

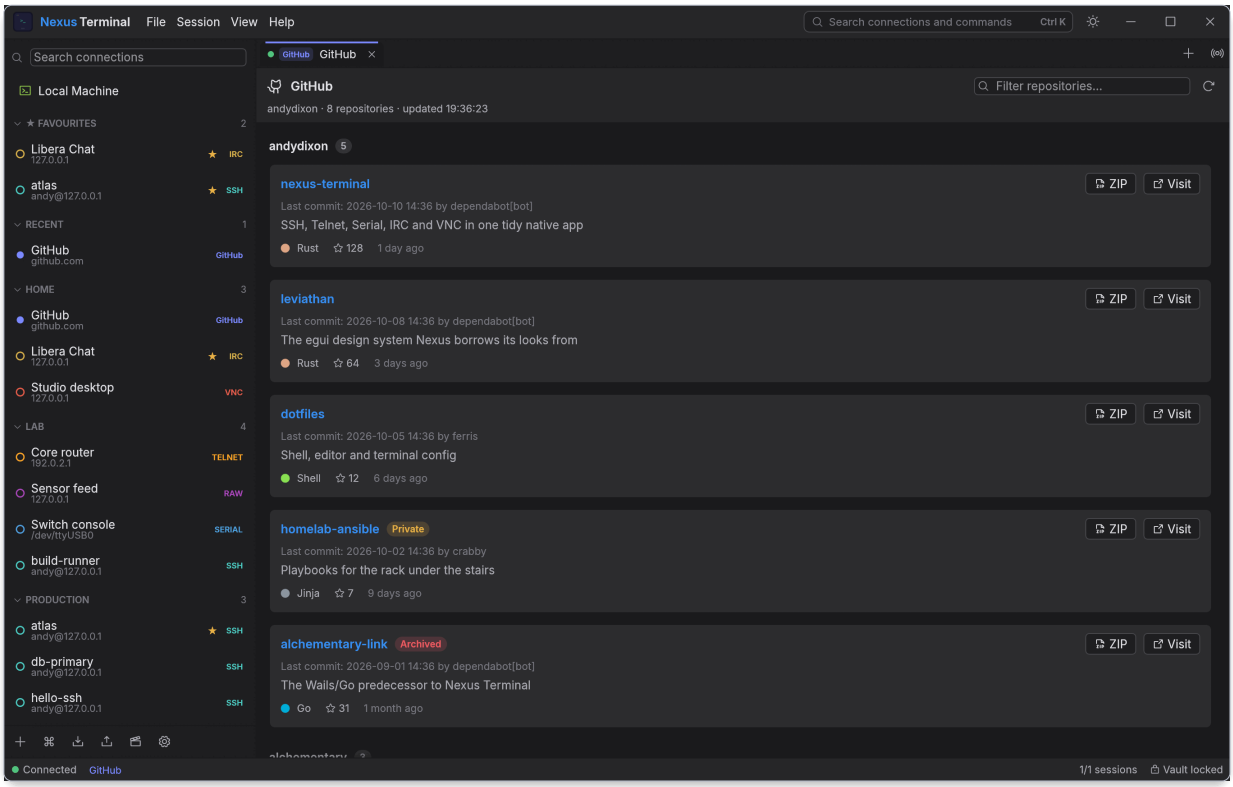
about **2 minutes** You'll see all your repositories, freshest first



A GitHub profile: just the API address and a token.

TL;DR

Paste a personal access token with the `repo` scope. Get a tidy, filterable list of every repository you can see.



Repositories grouped by owner, with the latest commit and who made it.

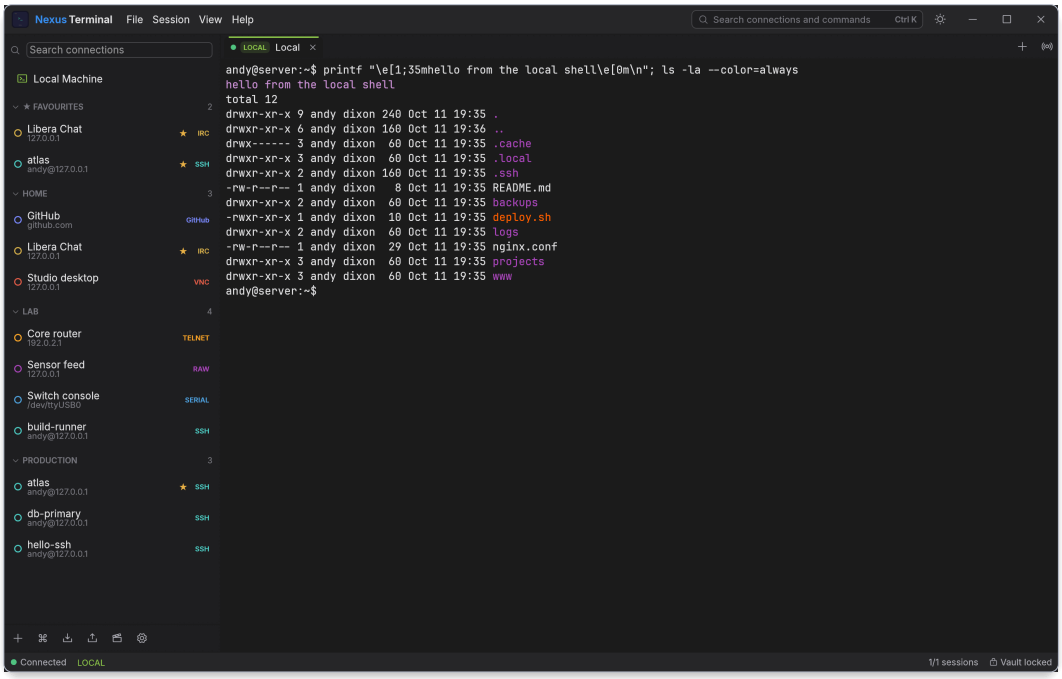
- Grouped by owner or organisation, sorted by most recent activity.
- Badges for Private, Fork and Archived, plus language and stars.
- Filter box, **Refresh now**, **Visit** (opens in your browser) and **ZIP** (downloads an archive).
- Works with GitHub Enterprise: set **API Base URL** to your server's API address. Refreshes every 60–3600 seconds.

CONNECT · CHAPTER 10

Your own computer

about **1 minute** | You'll stop alt-tabbing to a separate terminal app

Click **Local Machine** for a shell on this computer. It uses your system's default shell unless you choose another in **Settings** → **General** → **Local shell** (`pwsh`, `zsh`, `fish`, whatever you fancy).



A local shell, with all the same terminal powers as SSH.

3

PART 3

Work faster

Fewer clicks, less typing, more getting on with it.

- Tabs and split panes
- Terminal superpowers
- Never type it twice
- Files: SFTP and drag-and-drop
- Port forwards
- Recordings

Tabs and split panes

about **4 minutes** You'll juggle a dozen sessions without losing any of them

TL;DR

Right-click a tab for everything. `Ctrl / ⌘` + `Shift` + `D` splits right, `Ctrl / ⌘` + `Shift` + `E` splits down. `Ctrl / ⌘` + `1` to `Ctrl / ⌘` + `9` jump straight to a tab.

11.1 Tabs

Move Drag tabs about, or right-click → **Move Left** / **Move Right**.

Pin Right-click → **Pin**. Pinned tabs stay at the front.

Rename Right-click → **Rename**. "prod DB, DON'T TOUCH" is a perfectly valid tab name.

Duplicate Opens another session to the same profile.

Reconnect Does what it says.

Peek Hover a tab for its name, host, state and when it connected.

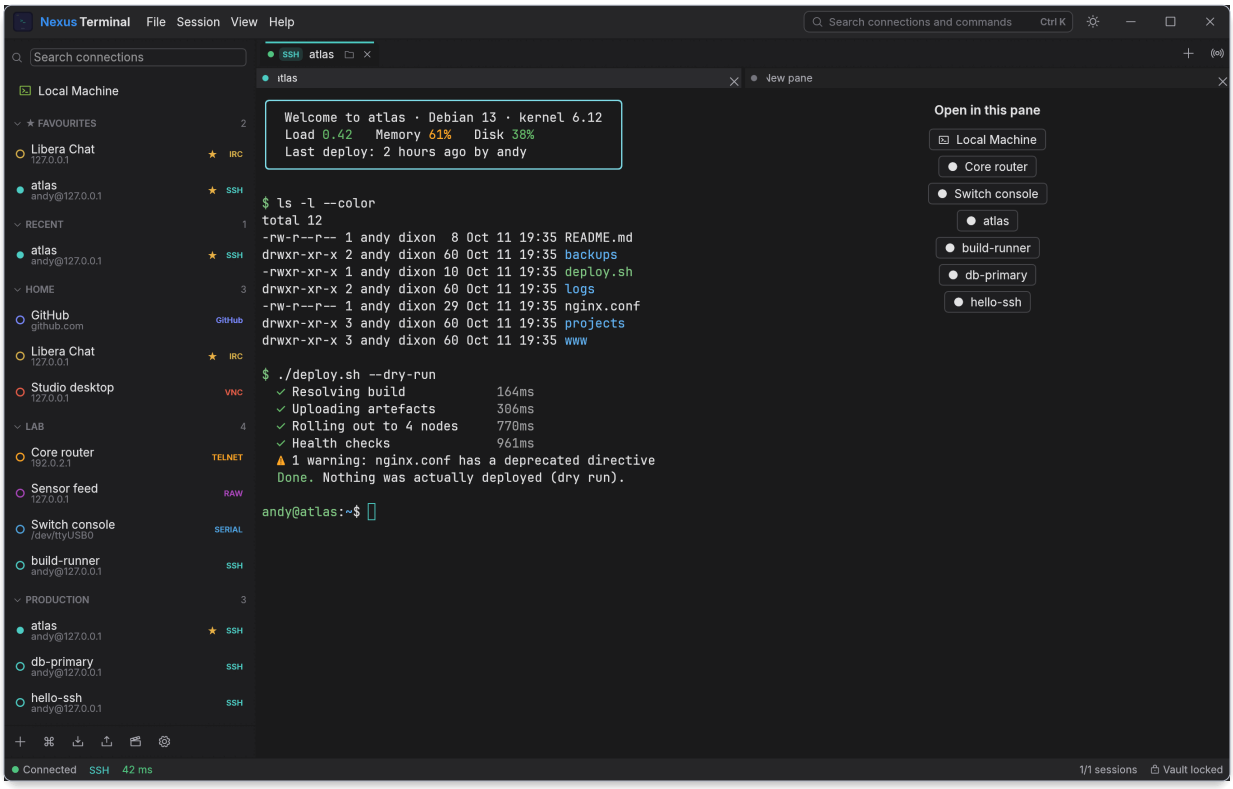
Close The ×, `Ctrl / ⌘` + `W`, or right-click → **Close**. Connected tabs ask first (you can turn that off in Settings).

Switch `Ctrl / ⌘` + `Tab` and `Ctrl / ⌘` + `Shift` + `Tab`, or `Ctrl / ⌘` + `1` to `Ctrl / ⌘` + `9`.

Lots of tabs? Scroll the tab strip with your mouse wheel or trackpad.

11.2 Split panes

Want two sessions side by side? Split the focused pane right with `Ctrl / ⌘` + `Shift` + `D` or down with `Ctrl / ⌘` + `Shift` + `E` (also in the **Session** menu, the right-click menu and the palette). The new pane asks what to open: **Local Machine** or any of your terminal profiles.



A fresh split. Pick what goes in the new pane.

Drag a divider to resize. Click **×** on a pane's header to close it and its neighbour takes the space. Closing the tab closes every pane in it.

> **Quick win**
>
> Split right, open `htop` on one side and tail a log on the other. Now you look like a hacker in a film.

Terminal superpowers

about **4 minutes** You'll copy, search and scroll like you've used it for years

TL;DR

Select to copy, `Ctrl / ⌘` + `F` to search, Ctrl/⌘-click links. Right-click for the rest.

12.1 Keyboard and mouse

Copy	Select text, then `Ctrl / ⌘` + `C`. Nothing selected? `Ctrl` + `C` interrupts the program as usual.
Paste	`Ctrl / ⌘` + `V`. Pastes are bracketed when the program asks for it, so editors won't auto-indent your paste into a staircase.
Select	Drag for a range, double-click a word, triple-click a line, `Shift`-click to extend.
Links	Hold `Ctrl` (`⌘`) and click a URL to open it in your browser.
Scroll	Mouse wheel, or `Shift` + `PageUp`, `Shift` + `PageDown`, `Shift` + `Home`, `Shift` + `End`.

Why does the scroll wheel move the cursor in less? · optional, skip it if you're in a hurry

Full-screen programs (`less`, `vim`, `tmux`, `byobu`) use the terminal's "alternate screen", where the wheel sends arrow keys instead of scrolling. Programs with mouse support (`htop`, `vim` with `mouse=a`, `tmux` mouse mode) get your clicks directly. That's normal terminal behaviour, not a bug.

12.2 The right-click menu

Copy, **Paste**, **Select All**, **Find**, **Send Snippet**, **Port Forwards** (SSH), **Split Right**, **Split Down**, **Export Output** (saves the whole scrollback as a text file), **Clear** and **Reset**.

12.3 Find

`Ctrl / ⌘` + `F` opens the find bar. Toggle **Aa** for case-sensitive and **.*** for regular expressions. `Enter` for the next match, `Shift` + `Enter` for the previous one, `Esc` to close. It shows "3 of 17" so you know where you are.

12.4 When the connection drops

The pane dims and shows a **Disconnected** card with **Reconnect** and **Close**. If auto-reconnect is on, it says **Reconnecting…** with a **Cancel** button, and a toast tells you which attempt it's on.

If the problem was a password or host key, you get a friendly dialog explaining what happened, with **Retry**, **Edit Profile**, **Known Hosts** (for host-key trouble) and **Close**.

Never type it twice

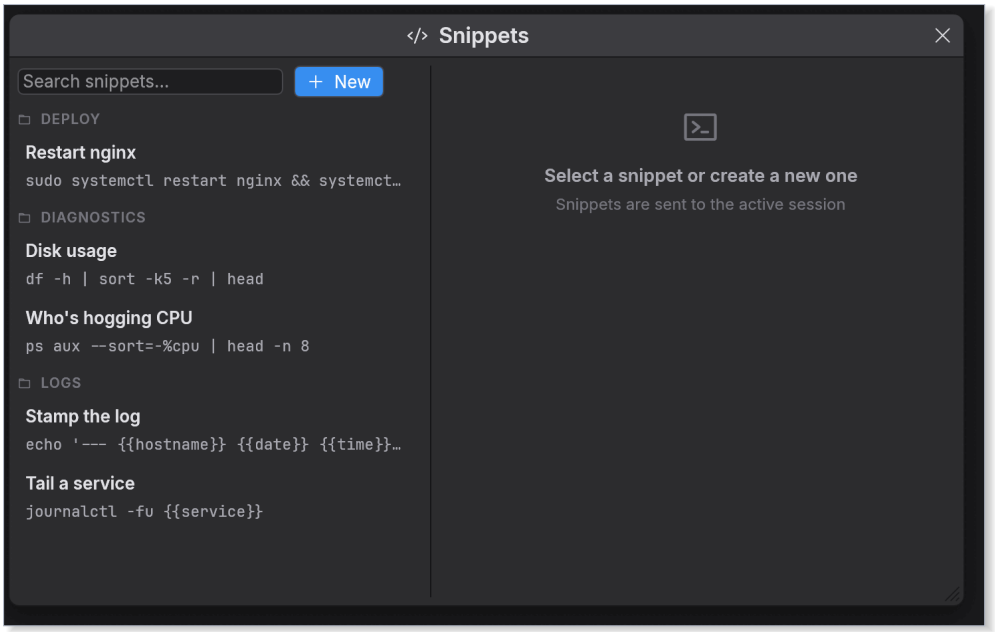
about **6 minutes** You'll stop retyping the same commands, and let Nexus Terminal watch the logs for you

TL;DR

Snippets store commands. **Broadcast** types into many sessions at once. **Triggers** watch output and tap you on the shoulder.

13.1 Snippets: `Ctrl / ⌘` + `Shift` + `S`

Save commands you use a lot, with a name, description and folder. Search them, click **Send to Session**, and the command is typed into the active session followed by `Enter`.



The snippet library, organised into folders.

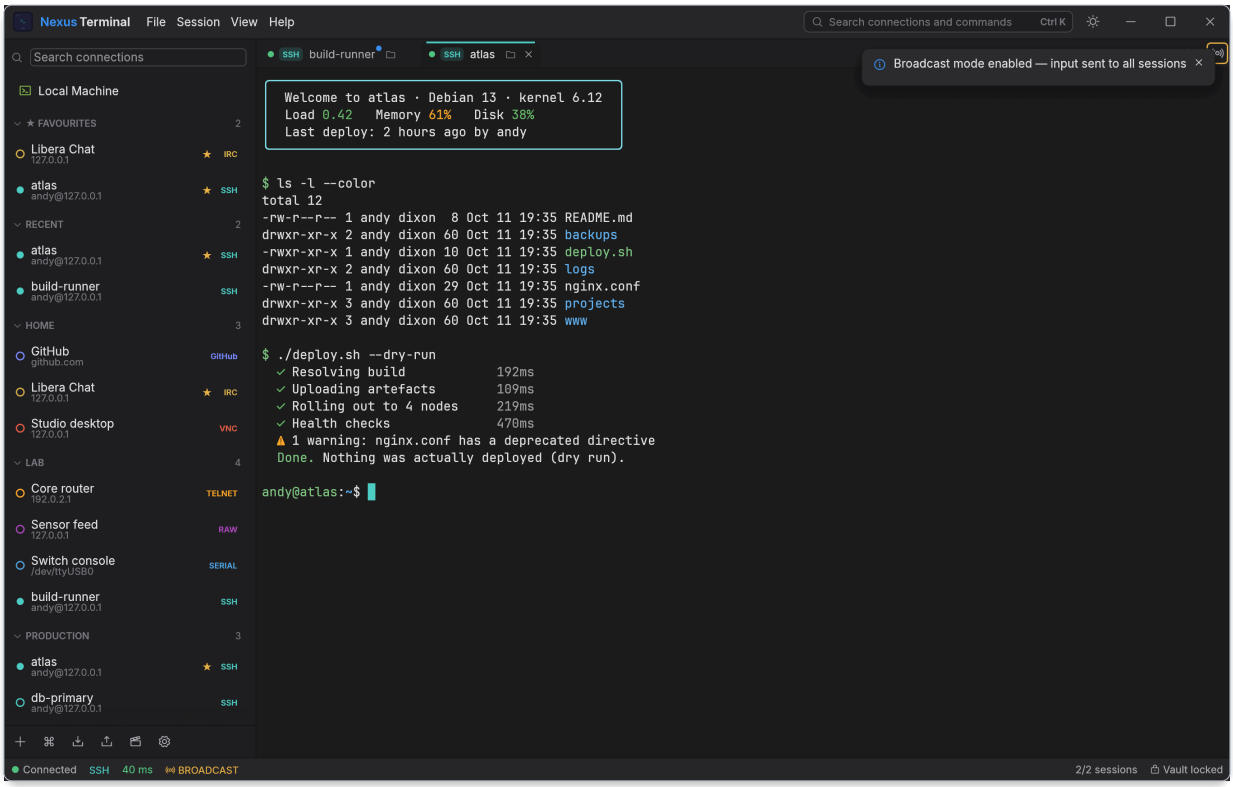
Snippets can have blanks to fill in:

- `{{hostname}}`, `{{date}}`, `{{time}}` and `{{timestamp}}` are filled in for you.
- Anything else, like `{{service}}`, is asked for when you send it.

So `journalctl -fu {{service}}` asks "service?" and tails whichever one you type.

13.2 Broadcast: `Ctrl / ⌘` + `Shift` + `B`

Turn broadcast on and everything you type goes to *every* connected terminal. The status bar shows **BROADCAST** so you can't forget it's on.



Broadcast is on: the status bar says so in no uncertain terms.

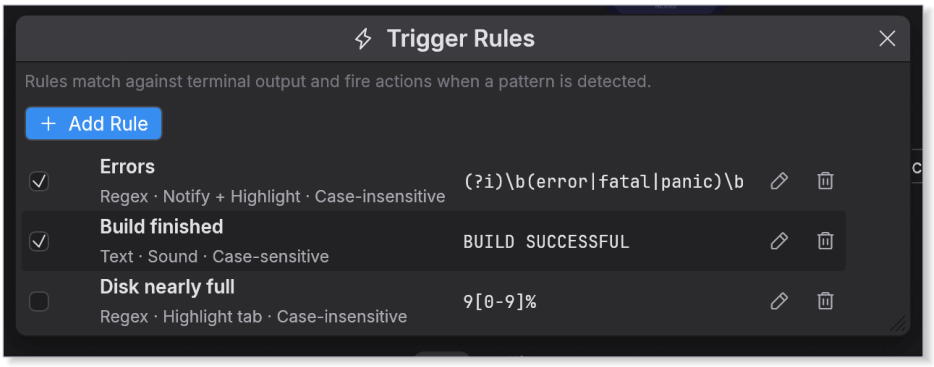
To broadcast to only some sessions, right-click the (((•))) button (or **Session** → **Broadcast targets…**), tick the ones you want and click **Apply & Enable**.

> **With great power…**
>
> Broadcast doesn't care which tab you *meant*. Double-check the target list before you type anything destructive. `sudo reboot` into twenty servers is very efficient and very memorable.

13.3 Triggers: let it watch for you

Session → **Trigger Rules** (or the palette). Each rule has a pattern (plain text or a regular expression), a case-sensitivity switch, an on/off switch, and an action:

Notify	shows a toast
Highlight tab	briefly outlines the pane
Notify + Highlight	both
Sound	asks your OS for attention (a bouncing dock icon or flashing taskbar button) and shows a toast



Three rules: shout about errors, ping when a build finishes, and one switched off for now.

Rules check each complete line of output, with colours stripped out first.

Quick win

Add a rule for `BUILD SUCCESSFUL` with the **Sound** action, start a long build, and go and make a cup of tea. Nexus Terminal will tell you when it's done. (Or `(?i)error|fatal` for the opposite.)

Files: SFTP and drag-and-drop

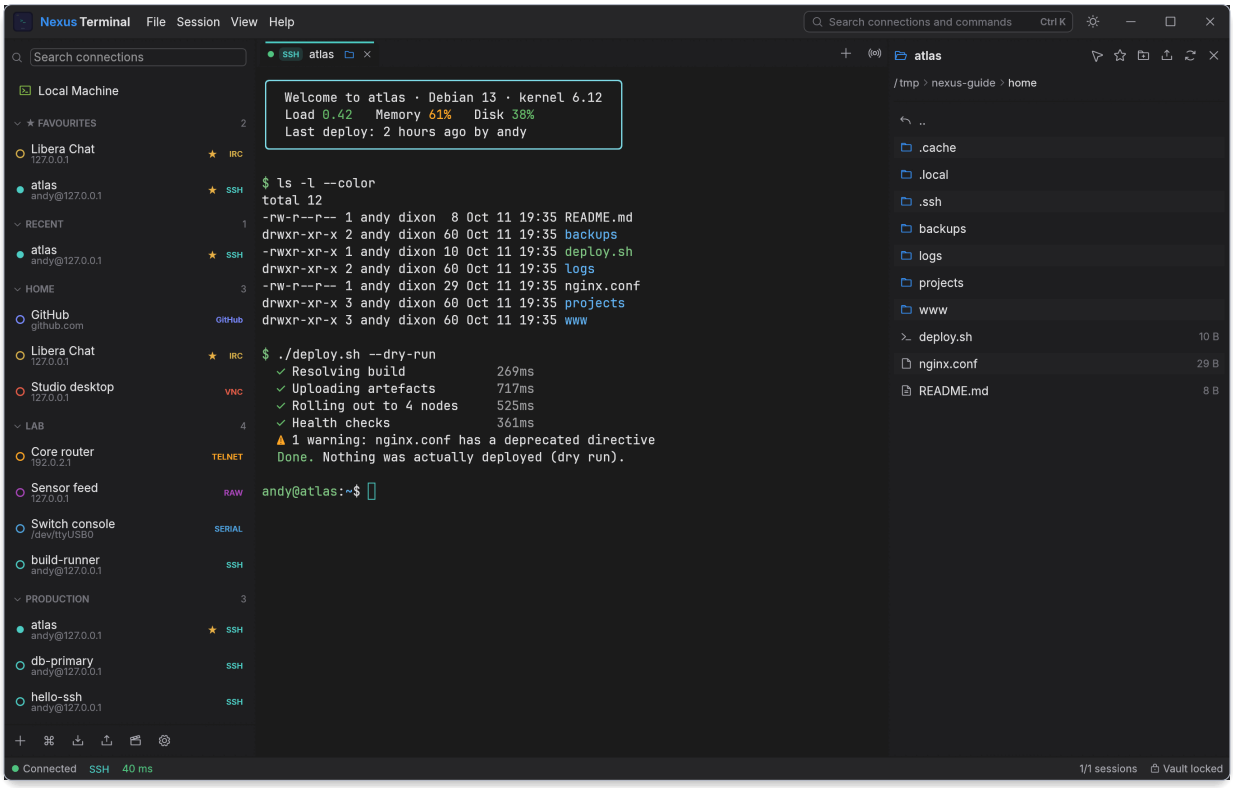
about **4 minutes** You'll move files to and from servers without leaving the terminal

TL;DR

Click the 🗁 on an SSH tab. Or drop a file onto the terminal and choose **Upload**.

14.1 The SFTP browser

Click the 🗁 folder icon on an SSH tab, **Session** → **SFTP browser**, or type "sftp" in the palette. It opens in the side dock next to your terminal.



Browsing the server's files next to the terminal.

- Move around with the breadcrumb trail, the `..` entry or **Go to path**.
- Upload files; download one or many to a folder you choose.
- Rename, delete (it asks first), make folders, refresh.

- ★ bookmarks the current folder, per profile.
- Ctrl-click (⌘-click) or Shift-click to select several things. Transfers show progress as they go.

14.2 Drag and drop

Drop files onto a terminal and choose:

Upload to current directory Pick the remote folder (SSH only).

Paste contents Types the file's contents into the terminal, as if you'd typed it.

The upload picker looks like the SFTP browser but shows folders only, labelled READ ONLY if you can't write there and NO READ if you can't look inside. Keyboard fans: ↑ ↓ to move, `Enter` to choose, `Backspace` to go up, `Esc` to cancel. If the SFTP browser is open, the picker starts in the same folder.

Port forwards

about **3 minutes** You'll reach a database or web app that only listens on the server

> **TL;DR**
>
> *Local* forwards bring a remote port to your machine. *Remote* forwards do the opposite.

Forwards saved in an SSH profile (under **Port Forwarding**, with **Active** ticked) start when you connect. To manage them on a live session, right-click the terminal → **Port Forwards**, or **Session** → **Port forwards**.

Local → Listen on your computer, forward to somewhere the server can reach. Example: `127.0.0.1:15432` → `127.0.0.1:5432` lets your desktop database app talk to Postgres on the server as if it were local.

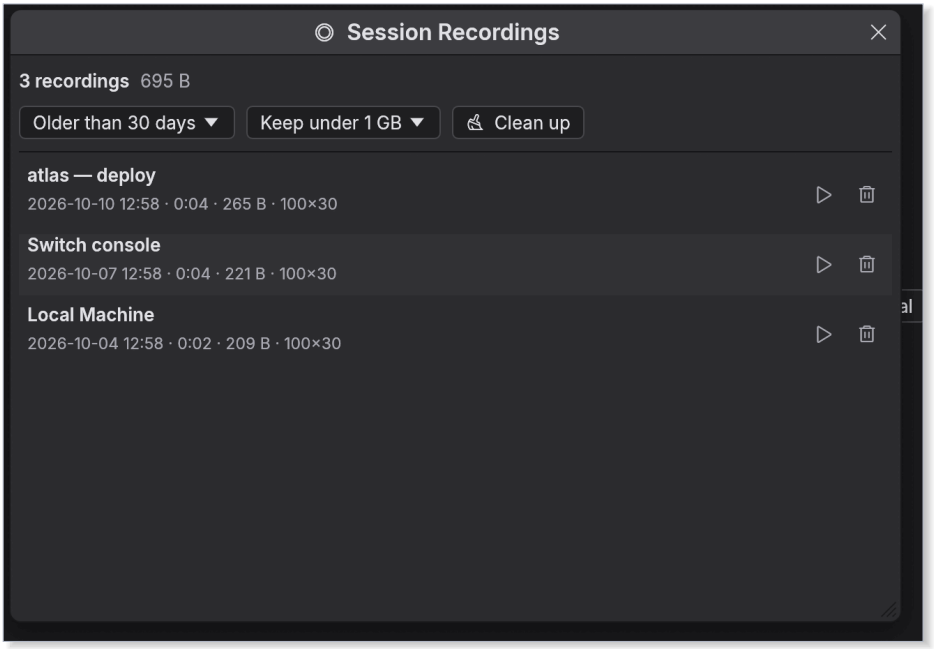
Remote ← Ask the server to listen and send connections back to somewhere *your* computer can reach. Great for showing a colleague the dev server running on your laptop.

Active forwards are listed with an L or R badge and a stop button. Toasts tell you when one starts, stops or fails.

Recordings

about **2 minutes** | You'll have a replayable record of exactly what happened

`Ctrl` / `⌘` + `Shift` + `R` starts and stops recording the active session. A red dot appears on the tab and **REC** in the status bar. Recordings are standard asciicast v2 files, so other tools (like `asciinema`) can play them too.



The recordings browser. Play, delete, or tidy up in bulk.

The browser (sidebar footer, **Session** menu or palette) shows how many recordings you have and how much space they take. Play or delete each one, or use **Clean up** to remove recordings older than 7, 30 or 90 days, or to keep the total under 1 GB or 5 GB. Playback has play/pause, restart, a seek bar and speed control.

> **Pro tip**
>
> Record the next time you do something fiddly and rare (a database failover, a certificate rotation). Six months later, "how did we do that again?" has an answer you can literally watch.

4

PART 4

Stay safe

Passwords, keys and host checks, minus the jargon headache.

- Passwords and the vault
- Host keys: is it really that server?
- SSH keys

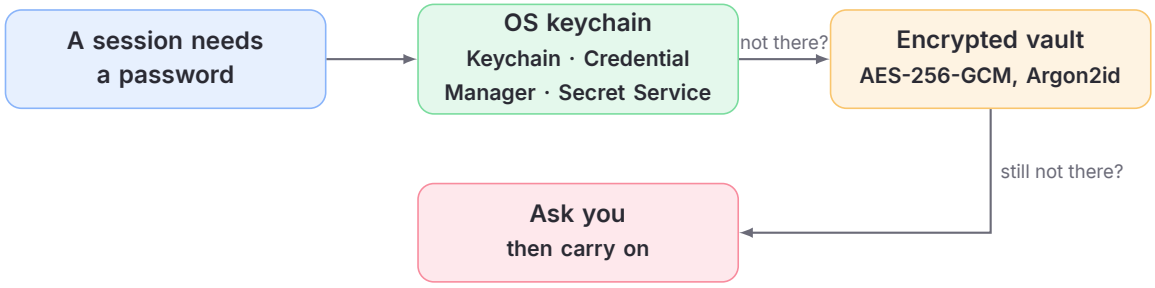
Passwords and the vault

about **5 minutes** You'll know exactly where your passwords live and who can read them

TL;DR

Passwords go in your OS keychain. If there isn't one, they go in an encrypted vault protected by a master password. For each password you choose: save it, remember it until you quit, or ask every time.

17.1 Where secrets live

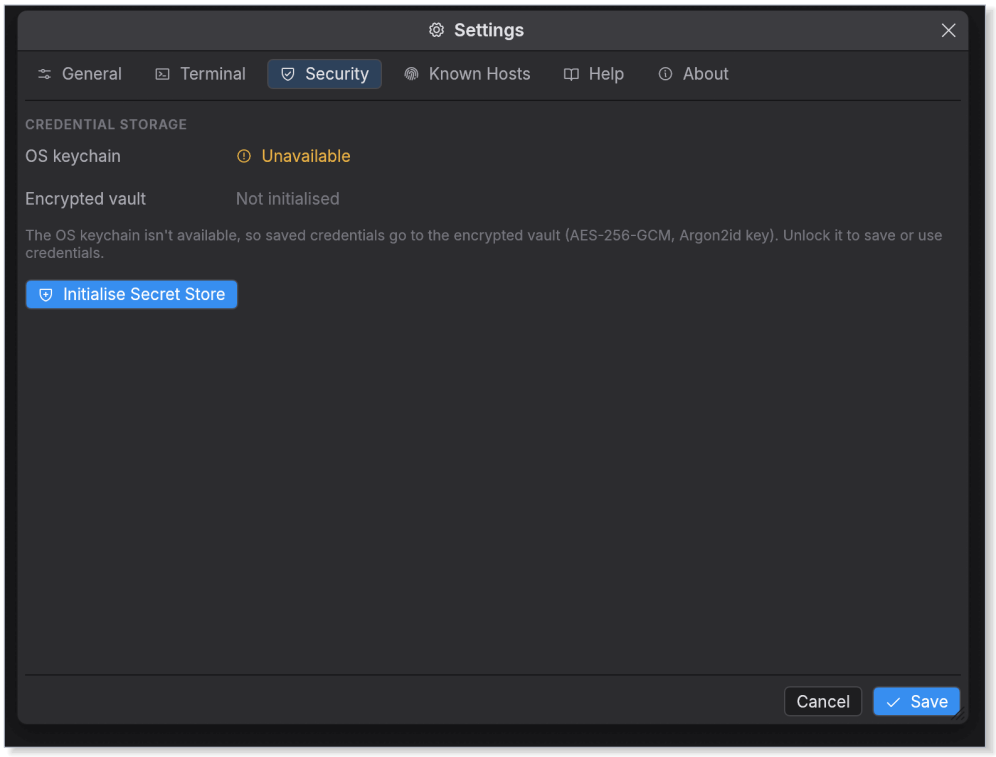


OS keychain first macOS Keychain, Windows Credential Manager, or the Linux Secret Service (GNOME Keyring, KWallet and friends).

Encrypted vault as backup If there's no keychain, or it refuses, secrets go into Nexus Terminal's own vault: AES-256-GCM with a key derived from your master password by Argon2id using 128 MiB of memory. In plain English: very, very expensive to brute-force.

17.2 Setting up the vault

Settings → **Security** → **Initialise Secret Store**. Choose a master password; the strength meter goes from Weak through Fair and Good to Strong. Unlock and lock it from the same page, from the palette, or by clicking the vault badge in the status bar. It locks itself when you quit.



Security settings on a machine without an OS keychain: the vault does the job instead.

> **There's no "forgot password" link**
>
> Lose the master password and the secrets in the vault are gone for good. That's the point of encryption. Put it in your password manager.

17.3 Three ways to handle each password

Every password, passphrase, token or TOTP secret has its own setting:

Save securely
Stored in the keychain or vault and used automatically. Most convenient.

Remember for this session
Asked once, kept in memory, never written to disk.

Ask every time
Asked on every single connection. Most paranoid.

When Nexus Terminal needs a secret it doesn't have, it asks. Press `Esc` or **Cancel** to give up on that connection.

> **What gets written down?** · optional, skip it if you're in a hurry
>
> Connections, disconnections, profile changes, credential access, imports and exports go into an audit log in the database. Secrets themselves never do. TOTP secrets live in the keychain or vault too, never in the profile.

Host keys: is it really that server?

about **3 minutes** You'll understand the scary fingerprint dialog and what to do about it

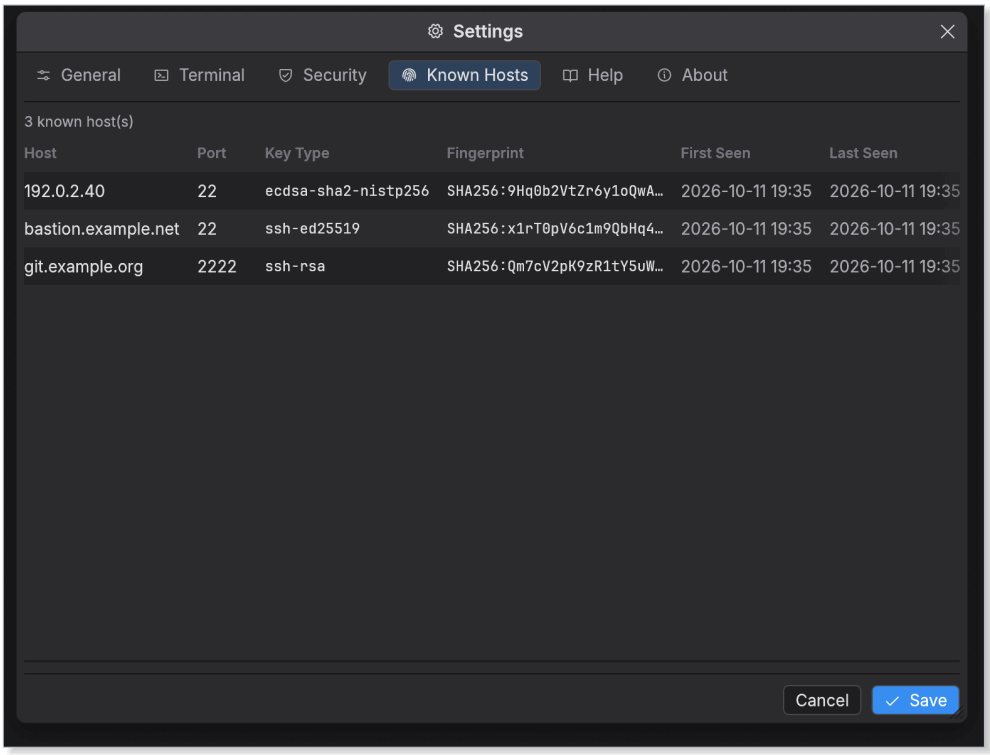
TL;DR

Every SSH server has a fingerprint. Nexus Terminal remembers it the first time and shouts if it ever changes. A changed fingerprint means a reinstalled server, or someone in the middle. Find out which before carrying on.

Pick a policy per SSH profile under **Host Verification**:

Strict (known hosts only)	Only connects to servers you've already trusted.
Ask on new or changed keys	Shows the fingerprint and asks. **Accept & Save** or **Reject**. The default.
Accept new, reject changed	Trusts first-time servers silently, still refuses changed keys.
Insecure (accept anything)	Exactly what it says. Lab use only.

Under every policy except *Insecure*, a key that differs from the saved one is rejected.



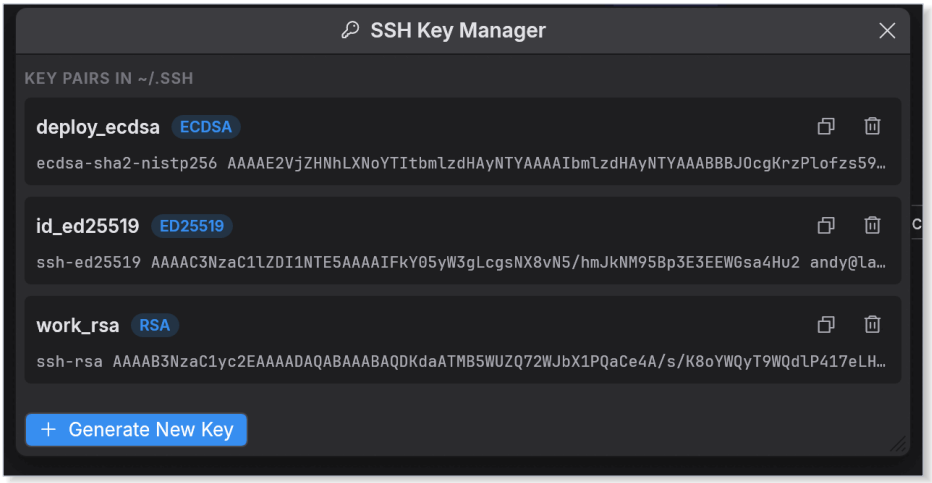
Known hosts. Click a fingerprint to copy it; **Remove** to forget a server.

> **Heads up**
>
> Nexus Terminal keeps its own list of known hosts. It doesn't read `~/.ssh/known_hosts`, so the first connection to each server asks once, even if your command-line `ssh` already knows it.

SSH keys

about **2 minutes** | You'll have a key pair without memorising ssh-keygen flags

Session → **SSH key manager** (or the palette) lists the key pairs in `~/.ssh` with their type. Copy a public key to the clipboard (ready to paste into GitHub or `authorized_keys`), delete a pair, or generate a new Ed25519, RSA (2048 or 4096 bits) or ECDSA key with an optional passphrase.



The SSH key manager.

Pro tip

Not sure which type? Pick Ed25519. It's small, fast and secure. Key generation uses your system's `ssh-keygen`, which comes with macOS, Linux and Windows 10 and later.

5

PART 5

Make it yours

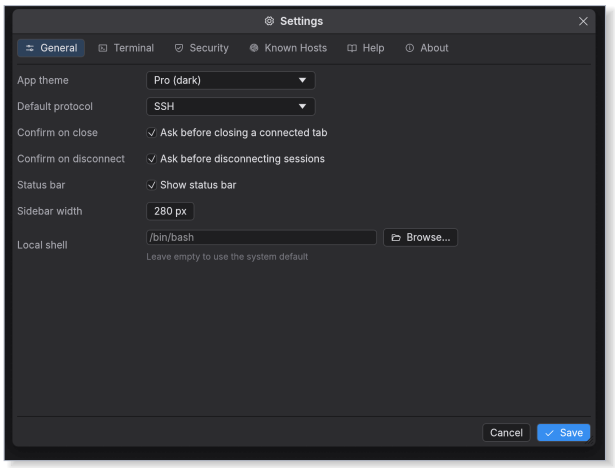
Settings, backups, and moving in from Alchementary Link.

- Settings
- Import and export
- Moving house

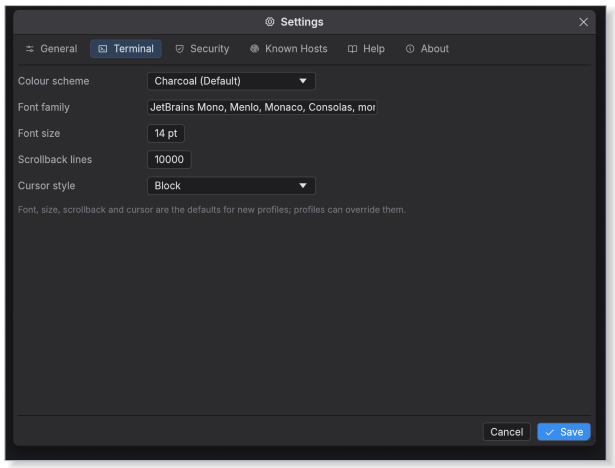
Settings

about **3 minutes** | You'll know where every setting lives

`Ctrl` / `⌘` + `,` opens settings (also the ⚙ in the sidebar footer and **File** → **Settings**). Change things, then click **Save**.



General



Terminal

General App theme (Pro, Midnight, Light), default protocol for new profiles, whether to ask before closing or disconnecting, status bar on or off, sidebar width, and the local shell (leave it empty for the system default, or **Browse...** for one).

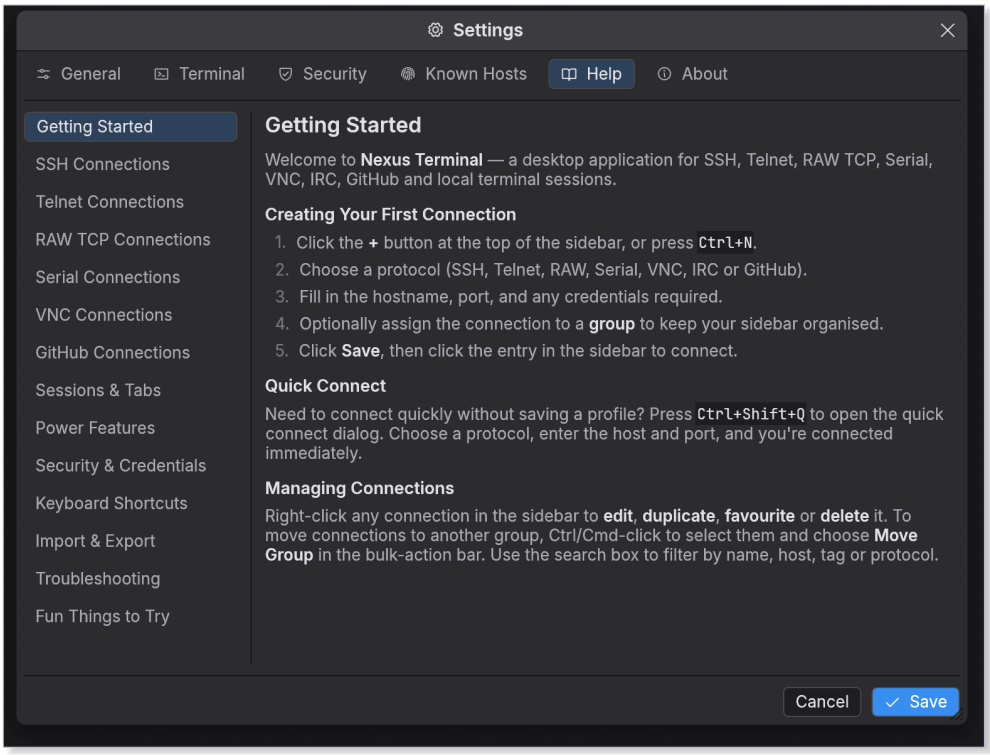
Terminal Colour scheme, font, size, scrollback and cursor style. These are the defaults for new profiles; each profile can override them.

Security Keychain and vault status; set up, unlock or lock the vault.

Known Hosts Saved server fingerprints, with first-seen and last-seen times.

Help Built-in help: connection types, shortcuts and troubleshooting, when this guide isn't to hand.

About Version, **Check for Updates** and **Import ~/.ssh/config**.



The built-in help, for when this guide isn't to hand.

20.1 Updates

Nexus Terminal checks once at launch. If there's a newer version the status bar says **Update available**; click it to see the release notes and a **View release** button. Nothing downloads or installs behind your back.

Import and export

about **3 minutes** You'll back up your profiles, or copy them to another computer

TL;DR

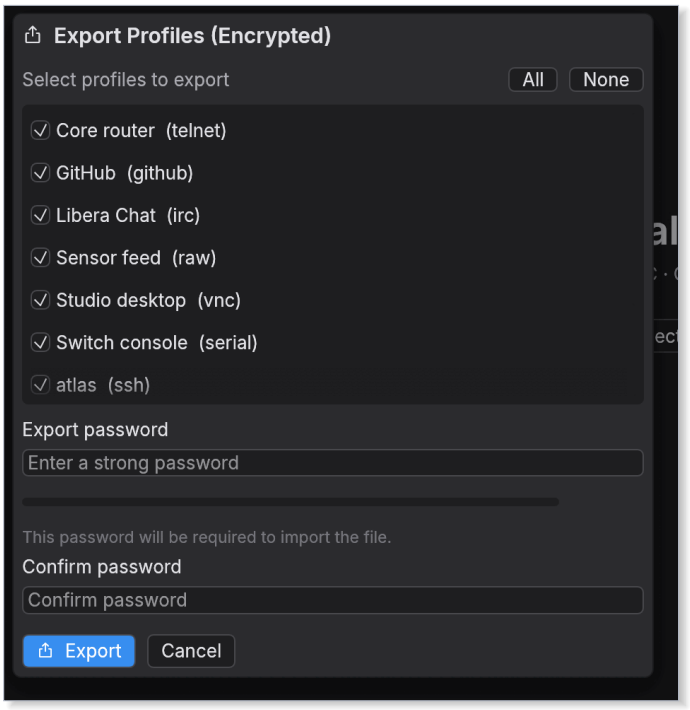
File → **Export profiles...** makes a password-protected `.nexus` file. **File** → **Import profiles...** reads it back. Already have an `~/.ssh/config`? Import that too.

Encrypted export Choose the profiles, set a password, save a `.nexus` file. Saved passwords aren't included, so the file is safe to email to yourself.

Encrypted import Pick a `.nexus` (or an old `.link`) file, enter its password and decide what happens to profiles that already exist: **Skip existing**, **Overwrite existing** or **Create duplicates**.

JSON **Export Profiles (JSON)** and **Import Profiles (JSON)** are in the palette. Tick **Include secrets** to keep proxy passwords and TOTP secrets, which are otherwise left out.

OpenSSH config **File** → **Import ~/.ssh/config** turns your `Host` blocks into profiles, with host names, users, ports, identity files and jump hosts.



Exporting profiles. Everything's ticked to start with; untick what you don't want.

JSON with secrets is not encrypted

A JSON export with **Include secrets** ticked has those secrets in plain text. Use the encrypted `.nexus` export to move things between machines.

Moving house

about **2 minutes** | You'll know what came over from Alchementary Link and what didn't

TL;DR

It's automatic. First run copies your Alchementary Link profiles, vault and recordings. Your old folder is left alone. Recreate trigger rules and SFTP bookmarks by hand.

When Nexus Terminal starts for the first time and finds an `AlchementaryLink` folder next to where its own data goes, it *copies* `link.db` to `nexus.db`, along with the `vault/` and `recordings/` folders, and shows a toast saying where they came from.

- ✓ Profiles, groups, favourites, known hosts, snippets and preferences
- ✓ Vault (same master password as before)
- ✓ Recordings
- ✓ Keychain passwords: picked up the first time each one is needed, then copied across
- ✓ Old `.link` export files still import
- ✗ Trigger rules and SFTP bookmarks (they lived in the old app's web storage)

Heads up

If a `NexusTerminal/nexus.db` already exists (say, from before the Alchementary Link rebrand), it's used as it is and nothing is copied.

22.1 Where everything lives

macOS
`~/Library/Application Support/NexusTerminal`

Windows
`%APPDATA%\NexusTerminal`

Linux
`$XDG_CONFIG_HOME/NexusTerminal`, usually `~/.config/NexusTerminal`

Inside: `nexus.db` (profiles, known hosts, preferences, snippets, IRC state and the audit log), `vault/vault.json`, `recordings/`, `triggers.json` and `sftp-bookmarks.json`. Keychain entries are stored under the service name `NexusTerminal`.

Try things without touching your real data

Start Nexus Terminal with `-data-dir /some/other/folder` and it uses that folder instead. Brilliant for experiments, demos and a portable install on a USB stick.

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PART 6

When things go wrong

Symptom first, fix second. No essays.

Troubleshooting, by symptom

Troubleshooting, by symptom

about **as long as it takes** — You'll find your symptom and the fix in one place

TL;DR

Find the heading that matches what you're seeing. Try the fixes in order. Most problems are the network, the password, or the baud rate.

23.1 Connecting

"Authentication failed"

- Check the username, password, key path and the ticked **Auth Methods**.
- If the key has a passphrase, check how the passphrase is stored (**Passphrase Mode**).
- Two-factor server? Click **Test** next to the TOTP secret and compare with your phone. Check your clock.
- Saved password not being used? The vault might be locked: click **Vault locked** in the status bar.

"Host key verification failed"

The server's fingerprint changed since you last connected. That happens when a server is rebuilt, but it's also exactly what an attack looks like. **Check with whoever runs the server.** If it's legitimate: **Settings** → **Known Hosts**, **Remove** the entry, reconnect and accept the new key.

It just says "Connecting…" forever

- Click **Test** in the profile editor. If that fails too, it's the network, a firewall or the wrong port.
- Behind a corporate proxy? Set one up under **Proxy** (SSH profiles).
- Going through a bastion? Check each jump host on its own first.

The session keeps reconnecting

Check the network and the credentials, then the profile's auto-reconnect settings. Reconnection backs off (1 s, 2 s, 4 s… up to 30 s) and stops after the maximum number of attempts.

23.2 Serial

The port won't open

- Is the adapter plugged in? Hit refresh in the device list.
- **Permission denied** on Linux: add yourself to `dialout` (or `uucp`), then log out and back in.
- On a Mac, use `/dev/cu.*`, not `/dev/tty.*`.
- **Port is in use by another program**: close `screen`, `minicom`, the Arduino IDE serial monitor or whatever else has it open.

Gibberish on screen

The baud rate, parity or stop bits don't match the device. Check its manual.

23.3 Remote desktops

VNC won't connect

- VNC ports are 5900 plus the display number: `:0` is 5900, `:1` is 5901.
- Check the password and how it's stored.
- Using an SSH tunnel? Check the tunnel's host, user, login method and key, and the host key policy.
- Connecting to a Mac? Turn on password access in macOS Screen Sharing settings (see p. 20).

23.4 Files

Upload is greyed out or fails

The folder is probably READ ONLY for your user. Pick another, or check permissions on the server. Uploads need an SSH session; other protocols can't do it.

"NO READ" on a folder

You don't have permission to look inside. Try the parent folder, or ask the server's admin.

23.5 The terminal itself

The terminal is the wrong size

Resize the window or nudge a split divider; the terminal re-fits itself. Still wrong? Reconnect. Very old servers may prefer a different **Terminal Type** in the profile (try `vt100`).

The scroll wheel moves the cursor instead of scrolling

You're in a full-screen program (`less`, `vim`, `tmux`). That's how terminals work: the wheel sends arrow keys there. Quit the program, or use its own scrolling (in `tmux`: `Ctrl` + `B` then `[`).

Emoji are in black and white

Known limitation: colour emoji show as monochrome glyphs. Everything else renders normally.

Still stuck?

Start Nexus Terminal with `-data-dir` pointing at an empty folder and try the same connection there. If it works, something in your profile is off; duplicate the working one and compare.

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PART 7

The extras

The cheat sheet, the command line, and words you might trip over.

- Keyboard cheat sheet
- Command-line options
- Glossary

Keyboard cheat sheet

about **print it** — Stick this page next to your monitor

Everywhere

Command palette	`Ctrl` / `⌘` + `K`
New connection	`Ctrl` / `⌘` + `N`
Quick Connect	`Ctrl` / `⌘` + `Shift` + `Q`
Settings	`Ctrl` / `⌘` + `,`
Full screen	`F11`

Tabs & panes

Close tab	`Ctrl` / `⌘` + `W`
Next tab	`Ctrl` / `⌘` + `Tab`
Previous tab	`Ctrl` / `⌘` + `Shift` + `Tab`
Go to tab 1–9	`Ctrl` / `⌘` + `1` … `9`
Split right	`Ctrl` / `⌘` + `Shift` + `D`
Split down	`Ctrl` / `⌘` + `Shift` + `E`

Power tools

Broadcast on/off	`Ctrl` / `⌘` + `Shift` + `B`
Snippets	`Ctrl` / `⌘` + `Shift` + `S`
Record on/off	`Ctrl` / `⌘` + `Shift` + `R`

In the terminal

Copy selection	`Ctrl` / `⌘` + `C`
Paste	`Ctrl` / `⌘` + `V`
Find	`Ctrl` / `⌘` + `F`
Next / previous match	`Enter` / `Shift` + `Enter`
Scroll a page	`Shift` + `PageUp`
Top / bottom	`Shift` + `Home` / `End`
Open link	`Ctrl` / `⌘` + click

Mouse moves

Select word / line	double / triple-click
Extend selection	`Shift` + click
Select several profiles	`Ctrl` / `⌘` + click
Everything else	right-click it

Command-line options

about **1 minute** — You'll script, demo and test with a few handy flags

`-data-dir DIR`	Use another data folder (portable installs, demos, testing).
`-theme pro|midnight|light`	Start with this theme.
`-connect NAME`	Connect to a saved profile by name at startup. Repeat it for several.
`-open WINDOW`	Open a window at startup: `settings`, `settings:security`, `snippets`, `keys`, `recordings`, `palette`, `quick`…
`-version`	Print the version and exit.

How the screenshots in this guide were made · optional, skip it if you're in a hurry

Every screenshot here is the real app, driven by `docs/guide/tools/screenshots.sh`: it starts throwaway SSH, IRC, VNC and GitHub-ish servers, fills a demo data folder, and runs Nexus Terminal headless with `-screenshot`, `-do` (run a palette command) and `-answer` (fill a password prompt). Change the app, rerun the script, rebuild the guide.

Glossary

about **dip in** Words you might trip over, in one line each

Agent A background program holding your unlocked SSH keys, so you type the passphrase once.

Asciicast The file format for terminal recordings. Plays in Nexus Terminal and `asciinema`.

Bastion / jump host A server you go *through* to reach the ones behind it.

Baud rate How fast a serial line talks. Both ends must agree.

Broadcast Typing into several sessions at once.

DTR / RTS Serial control lines. Some boards use them for reset or flashing.

Fingerprint A short summary of a server's host key, used to spot imposters.

Host key A server's identity. Changes when it's rebuilt, or when someone's pretending.

Keychain Your operating system's password store.

Port forward A tunnel carrying one TCP port through SSH.

Profile A saved connection.

RAW TCP A plain socket: bytes in, bytes out, no terminal rules.

SFTP File transfer over SSH.

Snippet A saved command, optionally with blanks to fill in.

TOTP Time-based one-time passwords: the six digits from your authenticator app.

Trigger A rule that watches output and reacts to a pattern.

Vault Nexus Terminal's encrypted password store, for when there's no keychain.

VNC A way of seeing and controlling another computer's desktop.



That's the lot.

Go and connect to something.